ASSIGNMENT -2 -CO2

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COURSE CODE:	**ITA0329**
COURSE:	**MOBILE COMPUTING**

Q1 – Mobile Communication or Mobile Computing

1. Classification criteria

- User mobility: whether the user/device moves while using the service.
- Connection type: wired, fixed wireless, or cellular/wireless connection.
- Disconnection tolerance: whether work can continue without a network.
- Context awareness: whether location, time or surrounding conditions affect the application.
- Computing capability: whether the device stores, processes and synchronises data locally.

Wireless means that data travels without a physical cable. Mobile means that the device or user can change location while remaining useful. Therefore, wireless and mobile are not identical. Mobile communication transfers information while a user/device moves; mobile computing includes processing, storing and using information on a portable device.

2. Classification of the five cases

Case	Classification	Main problem	Reason
(a) Rugged tablet using cellular work orders	Mobile	Both; mainly mobile computing	Staff move between sites; cellular link supports communication, while the app must handle changing coverage and work-order data.
(b) Laptop connected to Ethernet at offices	Nomadic / fixed-wired at each office	Mainly mobile computing	The laptop is carried between locations, but during use it has a wired connection. It is not continuously mobile over the network.
(c) Fixed wireless router at substation	Fixed wireless	Mainly mobile communication / networking	The endpoint is fixed, but the last connection is wireless. There is no user mobility.

(d) Push-to-talk handset	Mobile	Mainly mobile communication	The handset supports real-time voice communication while the user moves.
(e) Offline meter-reading tablet	Mobile	Mainly mobile computing	It records data locally and synchronises later. Local storage and reliable synchronisation are essential.

3. Additional ambiguous case

A delivery van has a laptop connected to a Wi-Fi hotspot inside the vehicle. The van moves between towns, but the laptop may be used only when the van is parked. It is mobile as a device and uses wireless access, but the application may behave like a nomadic system if it is used only at stops. This case shows that mobility depends on both device movement and how the service is used.

4. Store-and-forward simulation

Assumption: 1,000 meter updates are generated. Each update is attempted once. Link availability is p. For immediate sending, an update is delivered only when the link is available. For store-and-forward, every update is stored and retried once per minute until the link is available. The expected additional delay is $(1-p)/p$ minutes, assuming independent availability at each retry.

Link availability	Immediate delivered	Immediate lost	Queued delivered eventually	Queued delayed	Mean delay of queued updates
90%	900	100	1,000	100	0.11 min
60%	600	400	1,000	400	0.67 min
30%	300	700	1,000	700	2.33 min

Interpretation: immediate sending loses more updates as coverage falls. Store-and-forward avoids permanent loss but introduces delay and requires local storage, retry logic, timestamps and duplicate protection. The table is an analytical expected-value simulation; a random simulation may vary slightly.

5. Conclusion

A wireless link alone does not solve mobility problems. Mobile computing also needs offline operation, local storage, power management, synchronisation, conflict handling, security and recovery after disconnection. Cases (a) and (e) need application-level changes, especially offline queues and synchronisation. Case (d) mainly needs communication continuity and handover support.

Detailed Classification and Store-and-Forward Analysis

6. Meaning of each classification criterion

The five criteria of Section 1 are used as questions that can be answered for any case. The table below states how each is judged, so that the classification in Section 2 can be repeated by another reader and reaches the same result.

Criterion	Question asked	How it is judged	Effect on the design
User mobility	Does the user or device change place while the service is being used?	Continuous, between stops only, or none	Decides whether handover and changing coverage must be handled
Connection type	Is the link wired, fixed wireless or cellular/wireless?	Physical medium at the moment of use	Decides which network problems (loss, delay, handover) exist
Disconnection tolerance	Can the work continue when the link is lost?	High, medium or low	Decides whether local storage and later synchronisation are needed
Context awareness	Do location, time or surroundings change what the application does?	Yes, partly or no	Decides whether sensors, GPS or site information are required
Computing capability	Does the device store, process and synchronise data itself?	High, medium or low	Decides whether the device is a computing node or only a terminal

7. Criteria matrix for the five cases

Applying the criteria to the five cases of Section 2 gives the following matrix. It explains why cases (a) and (e) are classified as mobile computing, case (d) as mobile communication and case (c) as fixed wireless.

Case	User mobility	Connection	Disconnection tolerance	Context awareness	Computing capability
(a) Rugged tablet, cellular work orders	Continuous between sites	Cellular	Medium: coverage changes	Yes: site and job location	High: work-order data
(b) Laptop on office Ethernet	Between offices only	Wired when in use	High: can work offline	Low	High
(c) Fixed wireless router at substation	None	Fixed wireless	Low: link must stay up	No	Low: relay only
(d) Push-to-talk handset	Continuous	Cellular / radio	Low: real-time voice	Partly: talk group, location	Low
(e) Offline meter-reading tablet	Continuous between meters	Intermittent	Very high: works offline	Yes: meter location	High: local storage and sync

The matrix shows that mobility alone does not decide the classification. Case (e) and case (d) both have continuous mobility, but case (e) depends mostly on local computing and synchronisation, whereas case (d) depends mostly on the quality and continuity of the radio link.

8. Further ambiguous cases

Section 3 gave the delivery van with a Wi-Fi hotspot. Three further cases show that the same reasoning applies to other situations:

Case	Mobile device?	Used while moving?	Best classification	Reason
Smart watch connected to a phone by Bluetooth	Yes	Yes	Mobile computing (device-level) with short-range wireless	The watch moves with the user and keeps data locally, but its network access depends on the phone.
Passenger laptop on train Wi-Fi	Yes	Yes	Mobile communication with a nomadic user	The user is seated but the whole network moves, so the backhaul link changes cell or operator.
Field laptop used offline and docked in the evening	Yes	Only between sites	Nomadic computing	All use is at stops, and synchronisation happens at the dock.

9. Store-and-forward: derivation and extended simulation

In the store-and-forward model of Section 4, an update that fails at its first attempt is retried once per minute. Let G be the number of failed attempts before the first success. Since each attempt succeeds independently with probability p, $P(G = g) = (1 - p)^{g} \cdot p$, which is a geometric distribution. Its mean is $E[G] = (1 - p)/p$, so the expected extra delay is $(1 - p)/p$ minutes, which is the formula used in Section 4. Two related results follow: the fraction of updates that are delayed at all is $1 - p$, and the mean delay of only the delayed updates is $1/p$ minutes (for example 1.11 minutes at $p = 90\%$).

To check the analytical table, a random simulation was run with a fixed seed (2026): 1,000 updates for each availability value, each update retried once per minute until the link was available.

Link avai lability	Immediate delivered (expected / simulated)	Delayed updates (simulated)	Mean delay, all updates (theory / simulated)	Mean delay of delayed updates	95th percentile delay	Longest delay
90%	900 / 886	114	0.11 / 0.13 min	1.15 min (theory 1.11)	1 min	3 min
60%	600 / 598	402	0.67 / 0.67 min	1.65 min (theory 1.67)	3 min	8 min
30%	300 / 307	693	2.33 / 2.30 min	3.31 min (theory 3.33)	8 min	20 min

The simulated values agree with the expected-value table within normal random variation (for example 886 against 900 immediate deliveries at $p = 90\%$). The results also show that the averages hide a long tail: at $p = 30\%$ the mean delay is about 2.3 minutes but the longest update waited 20 minutes. A real system must therefore store updates for the worst case and not only for the mean.

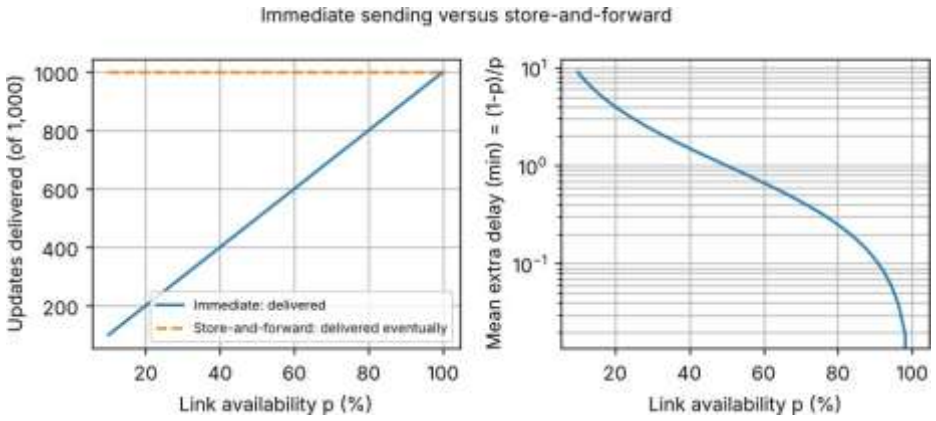


Figure Q1.1 – Updates delivered (left) and mean extra delay (right) as link availability changes.

10. Effect of the retry interval and of storage

The retry interval multiplies the delay: if the retry interval is t minutes the expected extra delay is $t(1 - p)/p$. For $p = 60\%$ the mean delay is 0.33 minutes for a 30-second retry, 0.67 minutes for a 1-minute retry and 3.33 minutes for a 5-minute retry. Shorter intervals reduce the delay but use more battery power because the radio wakes up more often, so the interval is a design trade-off.

Storage is also a requirement. If each meter update is 100 bytes (an assumed value), storing all 1,000 updates needs only 100 KB, which is easily available on a tablet. The storage must nevertheless survive power loss, so updates are written to non-volatile memory before the first send attempt. Each stored update should carry a sequence number and a time stamp, so that the server can remove duplicates when an acknowledgement is lost and can place late updates in the correct order.

11. Application-level design for cases (a) and (e)

Requirement	Case (a) work-order tablet	Case (e) offline meter tablet
Local storage	Cache of the assigned work orders and forms	Complete route and meter list for the day
Synchronisation trigger	Whenever coverage returns, in the background	When connected, or at the end of the route
Conflict handling	Server version wins for assignments; technician version wins for completion notes	Meter reading is append-only; duplicate readings are detected by meter ID and time stamp
Power management	Reduce polling in poor coverage	Switch the radio off while no network is available
Security	Encrypted storage and device lock; remote wipe	Encrypted storage; authenticated upload
Recovery after disconnection	Resume from the last acknowledged order	Retry from the oldest unacknowledged reading

12. Summary of the problems and solutions by case

Case	Main problem	Main solution
(a)	Changing coverage and work-order consistency	Offline queue, background synchronisation, conflict rules
(b)	Device moves but link is stable when used	Standard synchronisation at each office; no handover needed
(c)	Reliability of a fixed radio link	Link design, antenna placement and monitoring
(d)	Continuity of real-time voice	Handover support and coverage planning
(e)	Long periods without network	Local storage, store-and-forward and duplicate protection

13. A repeatable scoring procedure for the classification

To make the classification of Section 2 repeatable, each case can be scored from 0 to 2 on the five criteria and the scores added into two indices: a *mobile-communication index* (movement while in use, wireless link, need for real-time continuity) and a *mobile-computing index* (offline need, local data and processing, context awareness). The case belongs mainly to the class with the larger index. The scores are judgements, but writing them down shows how the decision was made.

Case	Movement in use	Wireless link	Real-time need	Comm. index	Offline need	Local data and processing	Context	Comp. index	Main class
(a)	2	2	1	5	1	2	2	5	Both
(b)	0	0	0	0	1	2	0	3	Computing (nomadic)
(c)	0	2	2	4	0	0	0	0	Communication
(d)	2	2	2	6	0	0	1	1	Communication
(e)	2	1	0	3	2	2	2	6	Computing

The scoring reproduces the classification of Section 2. Case (a) has equal indices (5 and 5), which is why it is classified as using both, with the application as the main problem. Cases (c) and (d) are dominated by the communication index and cases (b) and (e) by the computing index.

14. Problems that are specific to mobile computing

Problem	Cause	Effect on the application	Remedy
Variable bandwidth	Distance, interference and user load change the link rate	Slow responses; large transfers fail	Adaptive content size, compression, resumable transfers
Intermittent connectivity	Coverage holes, tunnels, handover	Lost requests; user waits	Offline queue, store-and-forward, retries with backoff
Limited battery	Radio and screen use energy	Short working time	Batch transfers, sleep modes, lower polling
Limited storage and processing	Portable size and cost	Cannot keep all data locally	Cache only the needed data; synchronise selectively
Security	Open radio medium; devices can be lost	Data exposure	Encryption, authentication, remote wipe
Location dependence	Services change with place	Wrong or stale data	Location-aware services; time stamps
Heterogeneity	Different networks and devices	Behaviour differs between users	Standard protocols, adaptive user interface

15. Mobile communication and mobile computing compared

Aspect	Mobile communication	Mobile computing
What is moved	Information between points while a user moves	Processing and data together with the user
Main concern	Coverage, handover, quality of the link	Offline operation, synchronisation, resource limits
Typical technologies	Cellular networks, radio links, push-to-talk	Mobile applications, local databases, synchronisation protocols
Failure seen by the user	Call dropped, poor voice quality	Data missing, conflicting updates, flat battery
Example in Q1	Cases (c) and (d)	Cases (a), (b) and (e)

16. Worked example: backlog during an outage

Suppose 1,000 meter tablets each generate one update every 15 minutes (an assumed rate) and the upload link can carry 100 updates per minute when it is working. The table shows how many updates accumulate during an outage and how long the queue needs to clear. The backlog during an outage of T minutes is 1,000 × T/15, and the clearing time is the backlog divided by the upload rate less the new arrivals (66.7 updates per minute arrive in total, so the spare rate is 100 − 66.7 = 33.3 per minute).

Outage (min)	Backlog (updates)	Storage at 100 bytes each	Time to clear the backlog
5	333	33 KB	10 min
30	2,000	200 KB	60 min
60	4,000	400 KB	120 min
120	8,000	800 KB	240 min

A 60-minute outage creates 4,000 queued updates (400 KB) that need 2 hours to clear. This shows the system must be designed for the recovery period and not only for the outage: if the upload rate were only 70 updates per minute, the spare rate would be 3.3 per minute and the queue would take about 20 hours to clear. The upload

capacity must therefore be well above the average arrival rate.

17. Review questions and short answers

Question	Short answer
Is a wireless device always mobile?	No. A fixed wireless router is wireless but its endpoint does not move (case c).
Is a mobile device always using wireless communication?	No. A laptop on Ethernet is carried between places but uses a wired link when in use (case b).
Why does store-and-forward avoid loss?	Updates are saved on the device and retried until the link is available.
What does store-and-forward cost?	Extra delay, local storage, retry logic, time stamps and duplicate protection.
Which cases need application changes?	Cases (a) and (e): offline queues and synchronisation.

18. Worked example of conflict handling

Case (a) uses a work order that the dispatcher and the technician can change. Suppose the work order is on the server with status “open” and priority “normal”. While the tablet is offline, the technician changes the status to “completed” and the dispatcher, at the office, changes the priority to “urgent”. When the tablet reconnects, the two changes are compared field by field with the version that both started from.

Field	Original value	Technician (offline)	Dispatcher (server)	Merged result	Rule
Status	open	completed	open (unchanged)	completed	Only one side changed it
Priority	normal	normal (unchanged)	urgent	urgent	Only one side changed it
Notes	empty	“part replaced”	empty	“part replaced”	Only one side changed it
Assigned crew	crew A	crew A	crew B	crew B	Server wins for assignments

Because the two users changed different fields, both changes are kept without asking anyone. A real conflict arises only if both changed the same field, for example the technician and the dispatcher both set the status to different values. In that case a rule decides (for example, the server value wins for assignments and the technician’s value wins for completion notes), and the losing change is stored in a history log so that nothing is lost silently.

19. Why duplicate protection is needed

If the tablet sends a meter reading and the acknowledgement is lost, the tablet cannot know whether the server received it and will send it again. Without protection the server would store two readings and the bill could be wrong. With a unique identifier for each reading (meter number plus time stamp plus sequence number) the server recognises the repeat and stores the reading only once, but still sends the acknowledgement so that the tablet can delete its copy. This is the reason why Section 4 lists duplicate protection together with timestamps among the requirements of store-and-forward.

Q2 – Sharing the Channel: FDMA vs TDMA

1. Capacity calculation

Given bandwidth per direction = 2 MHz. FDMA channel spacing = 25 kHz. TDMA carrier bandwidth = 200 kHz and each carrier has 8 time slots.

FDMA nominal channels = 2,000 kHz / 25 kHz = 80. TDMA carriers = 2,000 / 200 = 10; nominal slots = 10 × 8 = 80.

Assumptions for usable capacity: FDMA loses 10% of nominal capacity to guard-band and control allowance; TDMA reserves one of its eight slots per carrier for synchronisation/control. These are stated modelling assumptions, not universal standards values.

Scheme	Nominal capacity	Assumed overhead	Estimated usable capacity
FDMA	80 channels	10% guard/control allowance	72 channels
TDMA	80 slots	1 control slot per 8 slots on each carrier	70 traffic slots

2. Traffic comparison

Traffic type	Better expected utilisation	Reason
Continuous voice	FDMA	A continuous call occupies a stable frequency channel; there is little burst-idle time within an active call.
Bursty data (10% duty cycle)	TDMA, if slots can be scheduled/shared	A user transmits briefly and leaves other time available. Fixed dedicated slots can still waste capacity.
70% voice / 30% data	TDMA or hybrid	Voice can receive periodic slots while bursty data uses remaining slots dynamically. Traffic scheduling improves sharing.

3. Three designed traffic conditions

- Dataset A – Continuous voice: 60 users each hold a channel for a 3-minute call. Expected result: high occupancy in both systems; blocking occurs only when simultaneous calls exceed usable capacity.
- Dataset B – Bursty telemetry: 80 devices generate a short packet every 10 seconds and use about 10% duty cycle. Dedicated allocation wastes most reserved resources; pooled or dynamically assigned TDMA slots are more efficient.
- Dataset C – Edge case: 80 users generate packets at exactly the same instant (synchronised arrivals). A system with only 70 usable TDMA traffic slots must queue or block at least 10 simultaneous transmissions; FDMA with 72 free channels can serve 72 and block/queue 8.

For a dedicated-channel model, utilisation = busy resource time / total available resource time. Blocking = requests rejected because all resources are occupied. Wasted capacity = reserved but idle resource time. In the synchronised edge case, both systems experience a short burst of blocking; TDMA has 70 usable slots and FDMA has 72 assumed usable channels.

4. Design decision for campus telemetry

No. Dedicated FDMA/TDMA allocation is not justified when 90% of devices send only a 200-byte report per minute. The traffic is small and intermittent, so dedicated channels remain idle for most of each minute. A shared, scheduled packet channel with contention control or dynamic slot assignment is more suitable. Dedicated channels may be justified only where strict delay, guaranteed capacity or continuous traffic is required.

5. Conclusion

FDMA divides the frequency band; TDMA divides transmission time on carriers. Equal nominal capacity does not mean equal usable capacity because guard bands, synchronisation and control overhead differ. Traffic pattern determines actual utilisation.

Quantitative Comparison of FDMA and TDMA

6. Sensitivity of usable capacity to the overhead assumptions

The usable capacities of 72 FDMA channels and 70 TDMA traffic slots depend on the two overhead assumptions stated in Section 1. The tables below show how the result changes if the assumptions are changed, so that the comparison does not rely on a single number.

FDMA guard/control allowance	Usable channels (80 × (1 − allowance))
0%	80
5%	76
10% (assumed)	72
15%	68
20%	64

TDMA control slots per carrier	Traffic slots per carrier	Traffic slots on 10 carriers	Equivalent overhead
0	8	80	0%
1 (assumed)	7	70	12.5%
2	6	60	25%

One control slot per carrier equals an overhead of 12.5%, which is slightly higher than the 10% assumed for FDMA, so FDMA keeps a small advantage (72 against 70). If TDMA needed only one control slot for every two carriers, its usable capacity would be 75 slots and TDMA would be ahead. The ranking therefore depends on the overhead of each real system and not on the access method alone.

7. Quantifying the three traffic conditions

Using the usable capacities of 72 channels (FDMA) and 70 slots (TDMA), the three datasets of Section 3 give the following results.

Dataset	Demand	FDMA (72 channels)	TDMA (70 slots)	Comment
A – continuous voice	60 simultaneous calls	60 served, 83.3% occupancy, 0% blocked	60 served, 85.7% occupancy, 0% blocked	Both systems have spare capacity (12 and 10 units)
B – bursty telemetry, dedicated allocation	80 devices, 10% duty cycle	72 devices served, 8 without a channel (10%); reserved capacity only 10% busy	70 devices served, 10 without a slot (12.5%); 10% busy	Dedicated allocation fails to serve all devices although the average activity is only 8 devices
B – bursty telemetry, pooled slots	80 devices, 10% duty cycle	Not applicable	Mean 8 active devices; blocking is practically zero	Pooling needs far fewer resources than one per device
C – synchronised arrivals	80 simultaneous requests	72 served, 8 blocked or queued (10%)	70 served, 10 blocked or queued (12.5%)	Both systems block briefly; TDMA blocks two more

For dataset B with pooled slots, the number of active devices follows a binomial distribution with n = 80 and probability 0.1. The probability that more than 70 are active at the same instant is about 9×10^{-61}, so blocking is effectively impossible; even the 99.9th percentile of the activity is only about 17 devices. This explains why the pooled design needs a small fraction of the reserved resources.

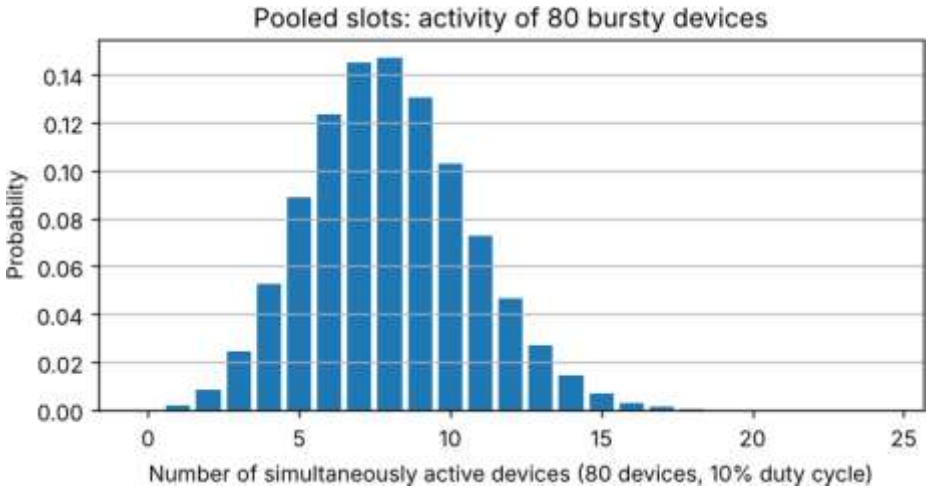


Figure Q2.1 – Probability that a given number of the 80 bursty devices are active at the same instant.

8. Blocking of continuous voice (Erlang-B)

For voice calls that are blocked when all resources are busy, the Erlang-B formula gives the blocking probability for an offered traffic A (in Erlangs) and m resources. The table compares 72 FDMA channels with 70 TDMA slots.

Offered traffic (E)	FDMA blocking (72 channels)	TDMA blocking (70 slots)
40	0.000%	0.000%
50	0.067%	0.137%
55	0.434%	0.742%
60	1.613%	2.374%
65	3.957%	5.223%
70	7.312%	8.957%
75	11.238%	13.098%
80	15.339%	17.291%

At 60 Erlangs, which is the load of dataset A if every user is active, FDMA blocks 1.6% of calls and TDMA blocks 2.4%. TDMA blocks more because it has two fewer usable resources. For a 2% target the supported traffic is about 61 E (FDMA) and about 58 E (TDMA).

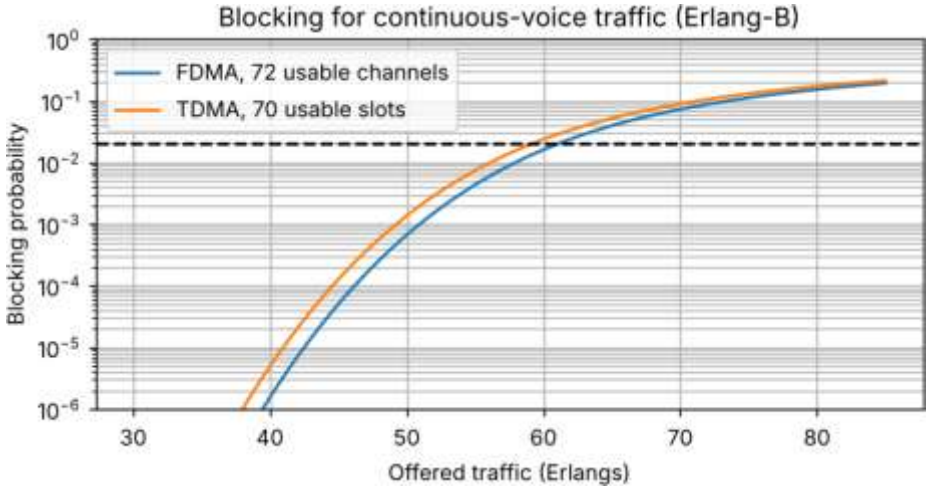


Figure Q2.2 – Blocking probability versus offered traffic for the two usable capacities; the dashed line is 2%.

9. Data-rate view of the campus telemetry decision

A 200-byte report once per minute is 1,600 bits per 60 seconds, an average of 26.7 bit/s per device. Assume (for illustration) that one channel or slot carries a net 10 kbit/s. A dedicated channel is then used for 0.27% of the time. If a shared channel is loaded to at most 50%, one channel can serve 10,000 × 0.5 / 26.7 ≈ 187 devices. The number of devices that 72 channels can serve is therefore:

Allocation	Devices per channel	Devices on 72 channels	Utilisation of each channel
Dedicated (one device per channel)	1	72	0.27%
Shared and scheduled (50% load)	about 187	about 13,500	50%

The 10 kbit/s value is an assumption. The conclusion does not depend on its exact value, because the ratio between the dedicated and the shared case is fixed by the duty cycle of the devices. This supports the decision of Section 4 that dedicated channels are not justified for the campus telemetry.

10. Hybrid FDMA/TDMA allocation

Real systems combine both methods: the band is divided into carriers (FDMA) and each carrier into time slots (TDMA). With the numbers of this question, 10 carriers of 200 kHz with 8 slots give the same 80 nominal channels as 80 carriers of 25 kHz. The hybrid has two advantages: the number of control and synchronisation channels is shared by several users, and a slot can be assigned to a user only while data is waiting. It also has costs: time synchronisation between users and the guard time between slots.

Property	FDMA	TDMA	Hybrid FDMA/TDMA
Resource unit	Frequency channel	Time slot on a carrier	Carrier and slot
Main overhead	Guard bands, control channel	Synchronisation, guard time, control slot	Both, but shared among users
Suited to	Continuous voice	Voice and scheduled data	Wide-area cellular systems
Weakness	Idle channel wastes bandwidth	Needs accurate timing	More complex planning

11. Capacity for other bandwidths

The calculation of Section 1 can be repeated for other bandwidths per direction, using the same carrier spacings and the same overhead assumptions (10% for FDMA and one control slot per carrier for TDMA).

Bandwidth per direction	FDMA channels (25 kHz)	FDMA usable (90%)	TDMA carriers (200 kHz)	TDMA nominal slots	TDMA traffic slots (7 of 8)
1 MHz	40	36	5	40	35
2 MHz	80	72	10	80	70
5 MHz	200	180	25	200	175
10 MHz	400	360	50	400	350

The ratio of the two usable capacities is constant ($0.9 \times 40/(0.875 \times 40) = 1.03$), so the 2 MHz result of the assignment scales directly to the other bandwidths. With a 10 MHz band the usable capacities are 360 channels and 350 slots, and the small FDMA advantage remains.

12. Spectral efficiency with an assumed voice rate

The efficiency of the two schemes can be compared as the information rate carried per hertz. Assume (for illustration only) that every voice channel or slot carries 13 kbit/s net.

Scheme	Usable resources	Net rate per resource	Total rate	Efficiency (bit/s/Hz)
FDMA	72	13 kbit/s	936 kbit/s	0.468
TDMA	70	13 kbit/s	910 kbit/s	0.455

Both values are far below the theoretical limit because of the modest voice rate and the overheads. The difference of 2.8% is the difference of the overhead assumptions and not a fundamental property, which supports the conclusion that equal nominal capacity does not give equal usable capacity.

13. Timing view of TDMA

In a TDMA carrier each user transmits only during its own slot, so the signal is a burst repeated once per frame. In a typical eight-slot system the slot lasts about 0.577 ms and the frame about 4.6 ms. A guard time at the end of each slot prevents bursts from different users, who are at different distances, from overlapping. The guard time wastes part of the slot, and the control slot per carrier assumed in Section 1 carries the synchronisation information that keeps all users aligned. FDMA does not need this timing, but needs guard bands in frequency and filters with sharp edges.

14. Utilisation as the duty cycle changes

The key difference between dedicated and pooled allocation is how the resources are used when devices are not always active. For 80 devices, the number of simultaneously active devices follows a binomial distribution. The table gives the number of resources that cover 99.9% of the instants for pooled slots, compared with 80 resources for dedicated allocation.

Duty cycle	Mean active devices	Dedicated resources	Utilisation of dedicated resources	Pooled resources for 99.9% coverage	Saving by pooling
100%	80.0	80	100%	80	0%
50%	40.0	80	50%	54	32%
10%	8.0	80	10%	17	79%
5%	4.0	80	5%	11	86%
1%	0.8	80	1%	5	94%

At 100% duty cycle (continuous voice) pooling saves nothing, which is why FDMA is competitive for voice. At 10% duty cycle the pooled design needs 17 resources instead of 80, and at 1% only 5, which gives the saving of 79% and 94%. These values are for a 99.9% coverage target; a stricter target needs a few more resources.

15. Decision summary

Condition	Recommended access	Reason
Continuous traffic, strict delay	Dedicated FDMA or TDMA	Guaranteed resource, no scheduling delay
Mixed voice and data	TDMA or hybrid	Periodic slots for voice, remaining slots shared for data
Sparse bursty traffic	Pooled or scheduled access	Dedicated resources would be idle more than 90% of the time
Synchronised bursts above capacity	Queueing plus pooled slots	Short bursts of blocking cannot be removed by the access method alone

16. A small Erlang-B example calculated by hand

The Erlang-B recursion $B(m) = A \cdot B(m-1)/(m + A \cdot B(m-1))$, with $B(0) = 1$, can be checked by hand for a very small system with $A = 2$ Erlangs: $B(1) = 2/(1 + 2) = 0.667$, $B(2) = 2 \times 0.667/(2 + 1.333) = 0.400$, $B(3) = 2 \times 0.4/(3 + 0.8) = 0.211$. Three channels therefore block 21% of calls at 2 Erlangs. The same recursion was used to obtain the blocking values of Section 8 for 72 and 70 resources.

17. Blocking at a fixed load for different usable capacities

Usable resources	Overhead assumption	Blocking at 60 E
80	No overhead	0.22%
76	FDMA 5% allowance	0.65%
72	FDMA 10% allowance (assumed)	1.61%
70	TDMA one control slot per carrier (assumed)	2.37%
68	FDMA 15% allowance	3.36%
64	FDMA 20% allowance	6.04%

At 60 E the blocking ranges from 0.22% when all 80 nominal resources are usable to 6.0% when only 64 are usable, so the overhead assumptions matter at least as much as the choice between FDMA and TDMA.

Q3 – CDMA: Orthogonality, Spreading and Near-Far Problem

1. DS-CDMA operation

In direct-sequence CDMA, each user multiplies each data bit by a unique spreading-code sequence. A bit is represented by N chips. The receiver multiplies the received chips by the intended user's code and sums them. A positive sum decides bit 1; a negative sum decides bit 0.

For a spreading length N, processing gain PG = N and PG(dB) = 10 log10(N).

Code length N	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

2. Predictions before execution

- Synchronous users with ideal orthogonal Walsh codes: cross-correlation is zero, so users ideally do not interfere. BER should remain very low as users increase, provided enough orthogonal codes exist and timing/power are controlled.
- Random non-orthogonal codes: cross-correlation is generally non-zero. Multiple users create interference, so BER is expected to increase as the number of users grows.
- Orthogonal codes with one user transmitting at 10 times the power: ideal synchronous orthogonality can still separate users mathematically, but practical timing errors, channel effects and imperfect power control allow the strong user to dominate weaker users.

3. Correlation and receiver

For two code vectors c_i and c_j, normalised correlation is $\rho_{ij} = (c_i \cdot c_j)/N$. Orthogonal codes have $\rho_{ij} = 0$ for $i \neq j$. A code correlated with itself has $\rho_{ii} = 1$. The receiver's despreading operation is a correlation with the wanted code.

Code length	Ideal orthogonal cross-correlation	Expected behaviour
8	0	Lower processing gain; fewer available orthogonal codes.
16	0	More spreading and more code choices than length 8.
64	0	Highest processing gain of these choices; longer chip sequence and more processing.

4. Why orthogonality can fail

Walsh-code orthogonality assumes chip-level synchronisation. If users arrive at different times, the receiver sees shifted code sequences whose cross-correlation may not be zero. Multipath propagation can also create delayed

copies. Unequal received powers create the near-far problem: a nearby strong transmitter can mask a distant weak transmitter.

Power control is essential because the base station should receive users at approximately controlled power levels. This reduces the chance that one strong user overwhelms weaker users.

5. Simultaneous-user estimate

For an ideal synchronous Walsh system, a length-N code family can provide at most N mutually orthogonal codes, with one code reserved for the receiver/reference or system use depending on the design. A practical safe estimate is fewer than N users because guard resources, pilots, imperfect synchronisation and interference reduce usable capacity. For a target BER of 10^{-3}, the actual supported number must be measured in the chosen noise, channel and power-control simulation; it cannot be fixed from code length alone.

6. Conclusion

Longer spreading codes increase processing gain and can improve interference resistance, but they do not remove the need for synchronisation and power control. Non-orthogonal codes and near-far conditions increase interference and bit errors.

Q3 (continued) – Simulation Results and Capacity of CDMA

7. Walsh codes and their correlation

Walsh codes of length $N = 2^m$ are the rows of the Hadamard matrix built by the rule $H_{2N} = [[H_N, H_N], [H_N, -H_N]]$, starting from $H_1 = [1]$. The rows of H_8 and H_{16} were generated by program and their normalised correlation matrices $\rho_{ij} = (c_i \cdot c_j)/N$ were computed. For N = 8, 16 and 64 every diagonal entry is 1 and every off-diagonal entry is exactly 0, which confirms the entries of the table in Section 3.

For example, the codes $c_1 = (+ + + + + + + +)$ and $c_2 = (+ - + - + - + -)$ have $c_1 \cdot c_2 = 1 - 1 + 1 - 1 + 1 - 1 + 1 - 1 = 0$. A receiver that despreads the sum of the two spread signals with c_1 obtains $N \cdot b_1 + 0 \cdot b_2$, so user 2 contributes nothing. With random codes the same inner product is a random number with mean 0 and standard deviation $\sqrt{N}$, which is why interference appears.

8. Simulation of bit error rate against the number of users

A Monte-Carlo simulation of DS-CDMA was run with binary data, BPSK, an additive white Gaussian noise channel and Eb/N0 = 6 dB. The settings were a fixed seed, 6,000 bits per point, user 1 as the wanted user and all users at equal power. Four conditions were simulated: Walsh codes with synchronous users, random binary codes, Walsh codes when each interferer has a random timing offset of several chips, and random codes without noise (so only multiple-access interference remains). The single-user BER at 6 dB is about 0.0024 for BPSK, and the simulated single-user values (about 0.002) agree.

Processing gain N = 8 (9.03 dB):

Users K	Walsh, synchronous	Random codes	Walsh, timing offsets	Random codes, no noise
1	0.0013	0.0018	0.0013	< 0.0002
2	0.0015	0.0075	0.0037	< 0.0002
4	0.0027	0.0357	0.0158	< 0.0002
8	0.0013	0.1497	0.0695	0.2013

Processing gain N = 16 (12.04 dB):

Users K	Walsh, synchronous	Random codes	Walsh, timing offsets	Random codes, no noise
1	0.0020	0.0013	0.0020	< 0.0002
2	0.0020	0.0038	0.0022	< 0.0002
4	0.0012	0.0143	0.0033	< 0.0002
8	0.0018	0.0847	0.0345	0.0890
16	0.0012	0.1450	0.1092	0.1590

Processing gain N = 64 (18.06 dB):

Users K	Walsh, synchronous	Random codes	Walsh, timing offsets	Random codes, no noise
1	0.0027	0.0013	0.0027	< 0.0002
2	0.0020	0.0052	0.0025	< 0.0002
16	0.0018	0.0378	0.0040	0.0095
32	0.0008	0.0885	0.0067	0.0647
64	0.0030	0.1393	0.1042	0.1243

The results confirm the predictions of Section 2. For synchronous Walsh codes the BER does not increase with the number of users and stays at the single-user noise level (about 0.001 to 0.003, which is within the statistical resolution of 6,000 bits). For random codes the BER rises steadily with K; for N = 16 and K = 16 it is 0.145 with noise and 0.159 without noise. When the Walsh users are not chip-synchronised the BER also rises (0.109 for N = 16 and K = 16), which shows that the orthogonality advantage is lost if timing is not controlled. A longer code reduces the interference: with 16 random-code users the BER with noise is 0.145 for N = 16 but only 0.038 for N = 64, because the same 16 users are now spread over four times as many chips.

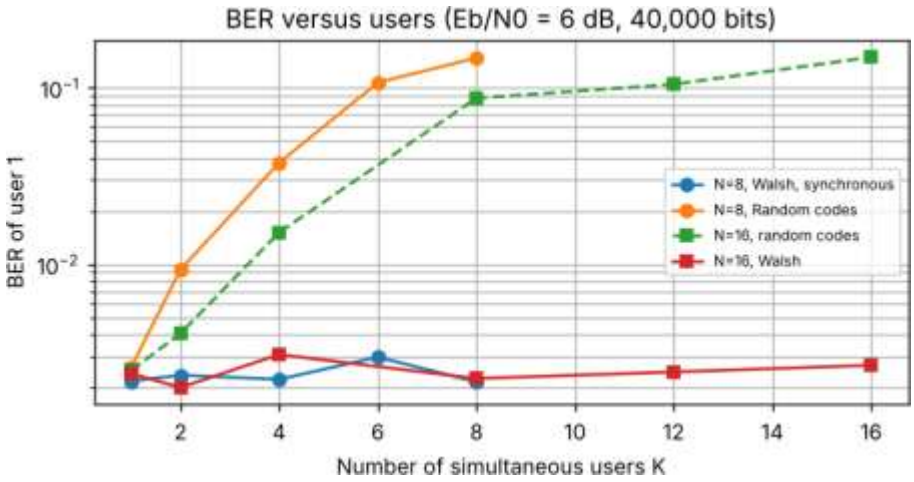


Figure Q3.1 – BER of user 1 versus the number of simultaneous users for synchronous Walsh codes and random codes (Eb/N0 = 6 dB).

9. Users supported at a target BER of 10^{-3}

Section 5 stated that the number of supported users cannot be fixed from the code length alone. A first estimate is possible for random codes with a Gaussian model of the interference: the despread interference variance is (K – 1)/N, so BER ≈ Q(√(N/(K − 1))). For BER = 10^{-3} the argument of Q must be 3.09, so N/(K − 1) ≥ 9.55 and K ≤ 1 + N/9.55 (before adding noise).

Code length N	Processing gain (dB)	Random codes: K at BER 10^{-3} (interference only)	Ideal synchronous Walsh codes
8	9.03	1	up to 8 (not more than N)
16	12.04	2	up to 16
64	18.06	7	up to 64

The comparison shows the value of orthogonality: with 64-chip random codes only about 7 users can be supported at BER 10^{-3}, whereas 64-chip Walsh codes can support up to 64 users if the system stays synchronised and power is controlled. In practice the Walsh limit is reduced by pilot and control codes, multipath and imperfect timing, so the safe number of users is lower than N, as stated in Section 5.

10. Near-far simulation

The near-far problem was tested with N = 16, four active users and Eb/N0 = 8 dB. One interfering user was given a power between 1 and 50 times that of the wanted (weak) user. Three cases were simulated: synchronous Walsh codes, Walsh codes with a one-chip timing offset on the interferers, and random codes.

Strong-user power / weak-user power	Walsh, synchronous	Walsh, 1-chip offset	Random codes
1×	0.0001	0.0008	0.0094
2×	0.0001	0.0012	0.0138
5×	0.0001	0.0029	0.0269
10×	0.0001	0.0070	0.0514
20×	0.0001	0.0191	0.1003
50×	0.0001	0.0892	0.2127

With perfect synchronisation the strong user has no effect (BER stays at 0.00013, the noise-only value). With a one-chip offset the weak user's BER rises from 0.0008 to 0.089 as the strong user's power increases to 50 times, a factor of more than 100. Random codes are worst: at equal power the BER is already about 0.009 and it reaches 0.21 at 50 times. At a 10× power ratio the random-code BER is about 5%, far above the 10^{-3} target. This confirms the prediction that the strong user dominates the weak one in practice, and it shows why CDMA base stations command mobile units to adjust their transmit power several times per second.

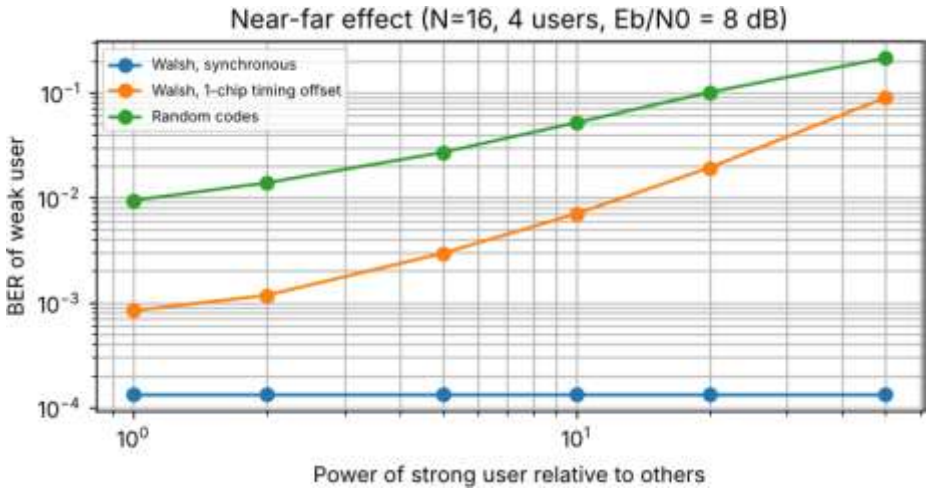


Figure Q3.2 – BER of the weak user as the power of the strong user increases (N = 16, four users, Eb/N0 = 8 dB).

11. Design guidance from the simulations

- Use the longest code that the bandwidth and processing power allow: each doubling of N adds 3 dB of processing gain.
- Treat Walsh orthogonality as an ideal: keep chip timing within a fraction of a chip and combine Walsh codes with a scrambling code when several cells are involved.
- Keep the received powers within about 1 dB of each other by fast closed-loop power control; a 10:1 imbalance already breaks the BER target for every code type tested.
- Reserve codes for pilot and control channels, and plan the user count below N.

12. Worked despreading example

Two users send one bit each with 8-chip Walsh codes: c_1 = (+1 +1 +1 +1 +1 +1 +1 +1) and c_2 = (+1 −1 +1 −1 +1 −1 +1 −1). User 1 sends b_1 = +1 (bit 1) and user 2 sends b_2 = −1 (bit 0). The transmitted chips are $b_1 c_1 + b_2 c_2$ = (0, 2, 0, 2, 0, 2, 0, 2). Despreading with c_1 gives 0 + 2 + 0 + 2 + 0 + 2 + 0 + 2 = +8, a positive sum, so the receiver decides bit 1. Despreading with c_2 gives 0 − 2 + 0 − 2 + 0 − 2 + 0 − 2 = −8, a negative sum, so the receiver decides bit 0. Both bits are recovered exactly, and the magnitude of 8 equals the spreading length N, which is the processing gain in linear form.

13. Effect of timing offsets on Walsh codes

Section 4 states that Walsh orthogonality needs chip-level synchronisation. To measure the effect, the cross-correlation between each Walsh code and every cyclic shift of every other Walsh code was computed. The table gives the average and the largest absolute normalised correlation, compared with random codes.

Length N	Synchronous Walsh	Shifted Walsh: average |ρ|	Shifted Walsh: largest |ρ|	Random codes: average |ρ|
8	0	0.136	1.00	0.274
16	0	0.089	1.00	0.194
64	0	0.039	1.00	0.099

The average interference of shifted Walsh codes is lower than that of random codes (for example 0.089 against 0.194 for N = 16), but the largest correlation is 1.0: for some shifts a Walsh code looks identical to another Walsh code. The worst case is therefore as bad as using the same code twice, which is why systems that cannot guarantee chip timing add a long scrambling code to randomise the interference.

14. Bandwidth, chip rate and jamming margin

Spreading multiplies the bandwidth by N. For a data rate of 9.6 kbit/s (a typical early cellular voice rate) the chip rate is N × 9.6 kchip/s:

Spreading length N	Processing gain (dB)	Chip rate	Approximate bandwidth
8	9.03	76.8 kchip/s	about 77 kHz
16	12.04	153.6 kchip/s	about 154 kHz
64	18.06	614.4 kchip/s	about 614 kHz
128	21.07	1.2288 Mchip/s	about 1.23 MHz

The jamming margin (the amount by which a narrowband jammer can exceed the wanted signal) is about the processing gain minus the required Eb/N0 and the implementation losses. For a required Eb/N0 of 7 dB (an assumed value) and 2 dB of losses, the margin is 18.06 − 9 = 9 dB for N = 64 and 21.07 − 9 = 12 dB for N = 128. This shows why the tactical link of Q6 uses a long spreading code.

15. Multipath and the RAKE receiver

Multipath creates delayed copies of the signal. For Walsh codes these copies are shifted versions, which are not orthogonal and add interference (Section 13). For spreading codes with good autocorrelation, however, each copy that is delayed by more than one chip can be separated and combined: a RAKE receiver uses several correlators, each aligned to one path, and adds the outputs. In this way multipath, which damages TDMA and FDMA, can be used to improve the signal in CDMA.

16. Power-control requirements from the simulations

Power imbalance (strong / weak)	Weak-user BER, 1-chip offset (N = 16)	Weak-user BER, random codes	Meets 10^{-3} target?
1×	0.0008	0.0094	Offset: yes; random: no
2×	0.0012	0.0138	No for random codes
5×	0.0029	0.0269	No
10×	0.0070	0.0514	No
50×	0.0892	0.2127	No

Even at equal power, random-code CDMA with four users does not meet the target of 10^{-3} at this Eb/N0, so a target BER requires both a longer code and a limited number of users. For the structured Walsh case the target is met only when the power imbalance is about one time (equal power) and the offset is small. Typical systems therefore command a power correction many times per second, so that the received powers stay within a small fraction of a decibel to a few decibels of each other.

17. Summary table

Question	Result of the study
Do orthogonal codes remove interference?	Yes, if users are chip-synchronised; BER stays at the noise level up to K = N
What happens with random codes?	BER grows with K; about 7 users can be supported at 10^{-3} for N = 64 (interference only)
What if timing is wrong?	Shifted Walsh codes can correlate fully; BER rises to about 0.11 for N = 16 and K = 16
What does a strong user do?	Raises the weak user's BER by more than 100× at 50× power (1-chip offset)

18. Chip timing in distance terms

Chip synchronisation can be expressed as a distance. At a chip rate of 1.2288 Mchip/s one chip lasts 0.814 µs, and radio waves travel 3×10^8 m/s × 0.814 µs ≈ 244 m in that time. Two users whose signals reach the base station with path differences of a large fraction of 244 m are therefore no longer aligned on the chip grid, and Walsh orthogonality is lost unless the mobile units adjust their transmit timing by the propagation delay (timing advance) or the spreading codes are scrambled. For a low chip rate such as 76.8 kchip/s the chip lasts 13 µs (3.9 km), so timing is much less critical, which is one reason why low-rate spreading is easier to synchronise.

19. Capacity concept compared with FDMA and TDMA

Property	FDMA / TDMA	CDMA
Limit on users	Hard limit: number of channels or slots	Soft limit: BER rises gradually as users are added
Effect of an extra user at the limit	Call blocked	All users see slightly more interference
Effect of silence in voice	Channel or slot remains reserved	Interference falls, so capacity rises
Frequency reuse between cells	Different channels in neighbouring cells	Same band in every cell, separated by codes
Main control need	Channel assignment and timing	Power control and code management

The soft-capacity property was seen in the simulation: BER rose smoothly with K for random codes (Section 8) rather than jumping at a fixed limit. The same property is useful when voice activity is about 40%, because users who are silent do not add interference and more users can be admitted. It also means that the planner must monitor BER and interference levels and not only count channels.

Q4 – OFDMA: Giving Each User the Subcarriers They Can Use Best

1. OFDM fundamentals

For useful symbol duration T_u, subcarrier spacing $\Delta f = 1/T_u$. This spacing makes subcarriers orthogonal over the useful symbol interval. A cyclic prefix (CP) copies the end of the OFDM symbol to its beginning. It reduces inter-symbol interference caused by multipath, provided the CP is longer than the channel delay spread.

Numerical orthogonality check: for subcarriers k and m over T_u, the inner product is $(1/T_u)\int_0^{T_u} \exp(j2\pi(k-m)t/T_u)dt$. It equals 1 when k=m and 0 when k≠m. Thus three different integer-indexed subcarriers are mutually orthogonal over the useful interval.

2. Allocation approaches

Approach	Method	Expected throughput	Fairness
Fixed contiguous blocks	Each of 8 users receives 8 neighbouring subcarriers.	Moderate; some users may receive poor-fading subcarriers.	Usually moderate; equal resource count, but not equal channel quality.
Interleaved round-robin	Users receive subcarriers spread across the band in a repeating pattern.	More robust to frequency-selective fading than contiguous blocks.	Generally good because resources are distributed across frequency.
Channel-aware best-subcarrier	Assign each subcarrier to the user with the best channel gain on it.	Highest sum throughput in many fading conditions.	Can be low; users with consistently weak channels may receive few resources.

A simple rate model is $R_{uk} = \log_2(1 + P_{uk}G_{uk}/N_0)$. Total throughput is the sum of allocated rates. Jain's fairness index is $J = (\Sigma R_u)^2 / (U \times \Sigma R_u^2)$, where U is the number of users. J=1 means equal rates; smaller values mean less fairness.

3. Prediction and design decision

Channel-aware allocation should give the highest sum throughput because it assigns each subcarrier to the user who can use it most efficiently. However, it may starve users with consistently weak channels. Interleaving is a useful compromise because each user samples different parts of the frequency band.

For a lecture hall where every student must receive a minimum rate, use proportional-fair or minimum-rate-constrained allocation: first guarantee the minimum rate to every active user, then allocate remaining subcarriers to users with good channel conditions. This protects weak users while still using good channels efficiently.

4. Conclusion

OFDMA divides the available subcarriers among users. Channel-aware assignment improves total throughput, while fairness-aware constraints prevent starvation. More users increase competition for subcarriers and can reduce the gain available to each user.

Numerical Study of OFDMA Allocation

5. OFDM parameters, cyclic-prefix overhead and the orthogonality check

The relation $\Delta f = 1/T_u$ links the subcarrier spacing to the useful symbol time. The table gives the numbers for two familiar systems and the cost of the cyclic prefix, which is the fraction $CP/(T_u + CP)$ of each symbol that carries no new data.

System (example)	Subcarrier spacing Δf	Useful symbol $T_u = 1/\Delta f$	Cyclic prefix	CP overhead	Longest delay spread tolerated
LTE-like, normal CP	15 kHz	66.7 µs	4.69 µs	6.6%	About 4.7 µs
LTE-like, extended CP	15 kHz	66.7 µs	16.7 µs	20.0%	About 16.7 µs
Wi-Fi 20 MHz, normal guard	312.5 kHz	3.2 µs	0.8 µs	20.0%	About 0.8 µs
Wi-Fi 20 MHz, short guard	312.5 kHz	3.2 µs	0.4 µs	11.1%	About 0.4 µs

A lecture hall has a very short delay spread (typically below 0.2 µs), so even the short guard interval is enough, whereas an urban cell with delay spreads of several microseconds needs the longer prefix. The prefix is a trade-off: a longer prefix protects against intersymbol interference but wastes more of the symbol time.

The orthogonality check of Section 1 was also carried out numerically by evaluating the inner product $(1/T_u)\int \exp(j2\pi(k - m)t/T_u)dt$ on 20,000 time samples for k, m = 1, 2, 3. The magnitudes obtained are:

	m = 1	m = 2	m = 3
k = 1	1.000	$< 1\times10^{-15}$	$< 1\times10^{-15}$
k = 2	$< 1\times10^{-15}$	1.000	$< 1\times10^{-15}$
k = 3	$< 1\times10^{-15}$	$< 1\times10^{-15}$	1.000

6. Simulation set-up for comparing the allocation approaches

The three approaches of Section 2 and a fourth, minimum-rate-guaranteed method were compared by simulation. The settings were: 8 users, 64 subcarriers, a frequency-selective Rayleigh channel with a six-tap exponentially decaying power profile (so neighbouring subcarriers are correlated), user average SNRs spread evenly from 0 dB to 20 dB (users at different distances), equal power on every subcarrier, the rate model $R_{uk} = \log_2(1 + P_{uk} G_{uk} / N_0)$, 1,500 independent channel draws and a fixed seed. The fourth method ("min-rate guaranteed") first gives each user its 4 best free subcarriers in random user order, and then assigns every remaining subcarrier to the user with the best gain on it.

Allocation method	Sum rate (bit/s/Hz summed over 64 subcarriers)	Jain's fairness index J	Mean rate of the weakest user
Fixed contiguous blocks	211.3	0.74	5.85
Interleaved round-robin	210.8	0.78	7.33
Channel-aware best subcarrier	422.7	0.22	0.00
Min-rate guaranteed (guarantee + best)	352.4	0.39	6.68

The results support the prediction of Section 3. Channel-aware allocation gives the highest sum rate (423), about twice that of the two fixed methods (211 and 211), but its fairness index is only 0.22 and the weakest user receives a mean rate of 0.0, which means that this user is starved. Interleaving gives the same total as contiguous blocks but a better fairness index (0.78 against 0.74) and a better rate for the weakest user (7.3 against 5.9), because each user samples subcarriers over the whole band. The min-rate method recovers most of the throughput (352, which is 83% of the best-subcarrier value) while keeping the weakest user at 6.7.

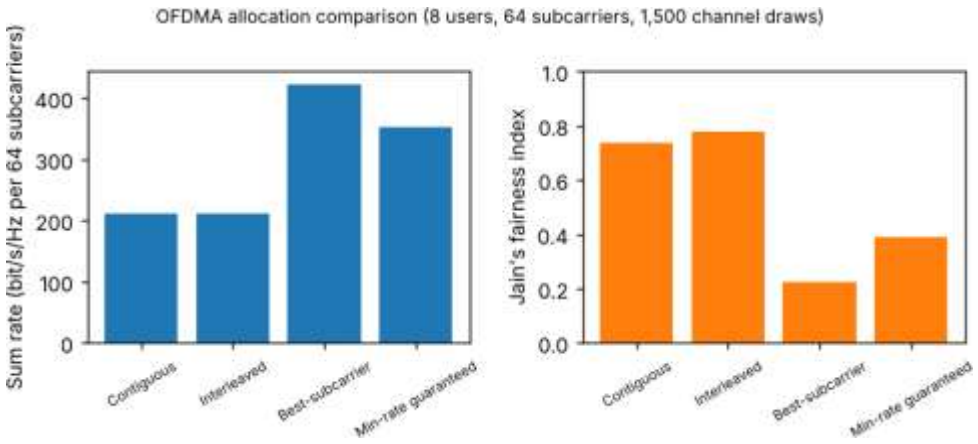


Figure Q4.1 – Sum rate (left) and Jain's fairness index (right) for the four allocation methods.

7. Effect of the number of users

The section 4 conclusion says that more users increase the competition for subcarriers. The simulation was repeated for 2 to 16 users with the same 64 subcarriers (500 channel draws each). With channel-aware allocation the sum rate increases with the number of users because each subcarrier has more candidates and the best one is chosen (multiuser diversity), whereas the interleaved allocation does not benefit and the rate available to each user falls.

Users U	Interleaved sum rate	Best-subcarrier sum rate	Gain of channel-aware method	Mean rate per user (best-subcarrier)
2	226	394	1.74×	197.2
4	214	405	1.89×	101.2
8	211	423	2.01×	52.8
12	202	434	2.15×	36.2
16	209	442	2.11×	27.6

The sum rate rises with U (from about 394 to 442) but the average rate per user falls from about 197 to 28, so the benefit of multiuser diversity is a gain for the cell and not for each individual user. This is consistent with the statement that more users can reduce the gain available to each user. In every run the user SNRs are spread evenly between 0 dB and 20 dB.

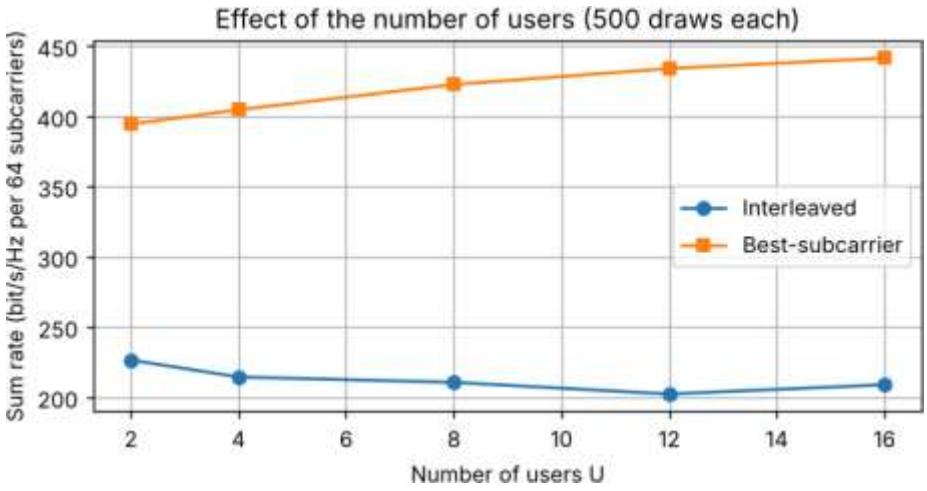


Figure Q4.2 – Sum rate versus number of users for interleaved and channel-aware allocation.

8. Algorithm for the lecture-hall decision

For the lecture hall, where every student must receive a minimum rate, the method used in the simulation can be written as the following procedure:

```
INPUT: U users, S subcarriers, channel gain G[u][k], minimum subcarriers m_min per user FOR each user u in random order:
        sort subcarriers by G[u][k] from best to worst
        assign the first m_min free subcarriers in this list to u FOR each remaining free
subcarrier k:
        assign k to the user with the largest G[u][k]
COMPUTE the rate of each user, the sum rate and Jain's index J OUTPUT: allocation map,
rates, J
```

The random user order prevents the same user from always choosing first. The parameter m_min sets the compromise: m_min = 0 gives the best-subcarrier method and m_min = S/U gives a nearly fixed allocation. In the simulation m_min = 4 (half of S/U = 8) was used. A rate guarantee in bit/s is obtained from m_min by choosing it so that m_min times the lowest expected per-subcarrier rate meets the requirement.

9. Limits of the model

- Equal power on every subcarrier is assumed; water-filling or adaptive power would change the rates.

- A fixed number of subcarriers per user is a simplification; real schedulers also allocate in resource blocks and use feedback that is delayed.
- Rates follow the Shannon-type formula, which is an upper bound; real modulation and coding give lower rates.
- The channel is assumed known; channel estimation errors reduce the gain of channel-aware allocation.

10. OFDM signal generation and reception

An OFDM transmitter maps the data to modulation symbols (for example QPSK or 64-QAM), places them on the subcarriers, converts them to the time domain with an inverse FFT, adds the cyclic prefix and transmits the result. The receiver removes the prefix, applies an FFT and obtains one complex value for each subcarrier, which only has to be divided by the channel gain of that subcarrier (one-tap equalisation). This is the main advantage of OFDM: the frequency-selective channel is converted into many flat sub-channels, which makes it easy to give different subcarriers to different users.

Step	Transmitter	Receiver
1	Map bits to symbols	Remove the cyclic prefix
2	Place symbols on subcarriers (user allocation decides which)	FFT to the frequency domain
3	Inverse FFT	One-tap equalisation per subcarrier
4	Add cyclic prefix	Pick the symbols of the allocated subcarriers
5	Transmit	Demap to bits

11. Resource units and raw rate

In an LTE-like system with 15 kHz spacing, a 20 MHz channel carries 1,200 active subcarriers and 14 OFDM symbols per millisecond. A resource block is 12 subcarriers over one slot of 7 symbols, so 100 resource blocks fill the channel. The raw symbol rate is $1{,}200 \times 14 \times 1{,}000 = 16.8$ million symbols per second, and with 64-QAM (6 bits per symbol) the raw bit rate before coding and overhead is 100.8 Mbit/s. In an OFDMA system these 100 blocks (or 1,200 subcarriers) are what the scheduler divides among the users.

12. A hand-worked allocation example

Three users share six subcarriers. The rates (bit/s/Hz) each user could achieve on each subcarrier are:

User	SC 1	SC 2	SC 3	SC 4	SC 5	SC 6
User 1	4.0	3.5	2.0	1.5	2.5	3.0
User 2	1.0	2.0	3.5	3.0	0.8	1.2
User 3	0.4	0.5	0.9	1.2	1.0	0.8

Method	Subcarrier owners (SC 1–6)	User rates	Sum rate	Jain index J
Contiguous blocks	1 1 2 2 3 3	7.5 / 6.5 / 1.8	15.8	0.82
Interleaved	1 2 3 1 2 3	5.5 / 2.8 / 1.7	10.0	0.81
Best subcarrier	1 1 2 2 1 1	13.0 / 6.5 / 0.0	19.5	0.60
Min-rate (each user takes its best free subcarrier first, then best)	1 1 2 3 1 1	13.0 / 3.5 / 1.2	17.7	0.57

The best-subcarrier method gives the largest sum rate, but user 3 receives nothing (rate 0) and J falls to about 0.60 (the minimum possible value for three users is 1/3 = 0.33). The min-rate method gives every user at least one subcarrier at a cost in the sum rate, and the interleaved and contiguous methods give intermediate totals with different fairness. The example shows the same trend as the simulation of Section 6, with numbers that can be checked by hand: $J = (\Sigma R)Z/(U \times \Sigma RZ)$.

13. Properties of Jain's fairness index

- J = 1 when all users receive the same rate.
- J = 1/U when one user receives everything (for U = 8 the lowest value is 0.125).
- J does not depend on the units of the rate or on the sum rate; two systems with different throughput can have the same J.
- J compares the users with each other and not with the rate each user needs, so it should be reported together with the minimum rate.

14. Lecture-hall scenario with assumed numbers

To see what the min-rate method means in practice, assume a lecture hall with 40 students sharing a 20 MHz OFDMA channel with 1,200 active subcarriers of 15 kHz (the LTE-like numbers of Section 11). Each student must receive at least 0.5 Mbit/s, and the average spectral efficiency on a good subcarrier is assumed to be 3 bit/s/Hz. The number of subcarriers a student needs is the required rate divided by the rate per subcarrier.

Quantity	Calculation	Value
Subcarriers per student if shared equally	1,200 / 40	30
Rate of one subcarrier at 3 bit/s/Hz	15 kHz × 3 bit/s/Hz	45 kbit/s
Subcarriers needed for 0.5 Mbit/s	500 kbit/s / 45 kbit/s	about 11.1, so 12
Subcarriers needed for all 40 students	40 × 12	480 (40% of the band)
Subcarriers left for channel-aware allocation	1,200 − 480	720 (60% of the band)

The guarantee uses 40% of the subcarriers, so 60% remain for the channel-aware allocation that raises the total throughput. If a student's channel is poor (for example 1.5 bit/s/Hz) the student needs 23 subcarriers for the same rate, and the guarantee for 40 such students (920 subcarriers) would leave only 280 subcarriers for the others. The required minimum rate therefore decides how much of the throughput gain of channel-aware allocation can be kept: the lower the guarantee, the closer the system is to the best-subcarrier method; the higher it is, the closer it is to a fair but less efficient allocation. This is the compromise that the design decision of Section 3 describes.

15. Summary of the numerical findings

Finding	Number
Highest sum rate (best subcarrier) compared with fixed allocation	About 2.0 times
Fairness of best-subcarrier allocation	J = 0.22; weakest user rate 0
Fairness of interleaving	J = 0.78
Throughput kept by the min-rate method	About 83% of the best-subcarrier sum rate
Cyclic-prefix cost (LTE-like normal CP / Wi-Fi normal guard)	6.6% / 20%

Q5 – SDMA: Reusing the Same Channel in Different Directions

1. Uniform linear array

For an N-element uniform linear array with element spacing d, wavelength λ and steering angle θ_0, the array factor can be written as $AF(\theta) = \Sigma_{n=0}^{N-1} \exp[jnkd(\sin\theta - \sin\theta_0)]$, where $k=2\pi/\lambda$. The normalised magnitude is $|\sin(N\psi/2)/(N \sin(\psi/2))|$, where $\psi=kd(\sin\theta-\sin\theta_0)$.

As N increases, the main beam becomes narrower and angular separation becomes easier. Side-lobe behaviour depends on array weighting; a uniformly weighted array has noticeable side lobes, so increasing N does not automatically eliminate them.

Array elements	Expected main-lobe width	Expected spatial separation
2	Wide	Poor separation
4	Narrower	Improved
8	Still narrower	Better for separated users
16	Narrowest of these arrays	Best angular discrimination of these choices

2. SIR criterion

Define the required SIR threshold as 10 dB. When the beam is steered toward user A, the desired signal is the array response in direction A and interference is the response in direction B. For equal transmit powers, $SIR(B) = |AF(\theta A)|^2 / |AF(\theta B)|^2$. The minimum angular separation is the smallest separation for which the calculated SIR meets 10 dB. The exact angle depends on element spacing, wavelength, steering angle and array weights; therefore it must be obtained from the pattern calculation rather than assumed.

3. Practical limitations

- Users close in angle produce similar spatial signatures, so the array cannot form two clean separate beams.
- Angular spread and multipath create several arrival directions for one user.
- A moving vehicle changes its direction and channel, requiring beam tracking.
- Calibration errors, mutual coupling and side lobes reduce practical SIR.

4. Toll-plaza decision

I would not rely on SDMA as the only access method at a toll plaza because most vehicles approach from nearly the same direction. Their angular separation may be too small for the array to distinguish them. I would combine SDMA with TDMA or another scheduling method, while using SDMA when vehicles are sufficiently separated.

5. Conclusion

SDMA reuses the same time and frequency resources by separating users spatially. Larger arrays create narrower beams, but clustered users and similar directions limit the achievable separation.

Q5 (continued) – Array Patterns, Separation and Toll-Plaza Geometry

6. Calculated beam properties of the uniform linear array

The array factor of Section 1 was evaluated numerically for N = 2, 4, 8 and 16 elements with spacing d = λ/2 and the beam steered to broadside ($\theta_0 = 0°$). The half-power beamwidth (HPBW) is the full width between the −3 dB points, the first null is the first angle at which the pattern reaches zero, and the side-lobe level is the strongest lobe outside the first null.

Elements N	Half-power beamwidth	First null (from the beam centre)	Highest side lobe	Minimum separation for SIR ≥ 10 dB
2	60.0°	87.1°	none (null at 90°)	52.7°
4	26.3°	30.0°	-11.3 dB	22.1°
8	12.8°	14.5°	-12.8 dB	10.7°
16	6.4°	7.2°	-13.1 dB	5.3°

The results agree with Section 1. Each doubling of N halves the beamwidth: 60.0°, 26.3°, 12.8° and 6.4°. The side-lobe level of a uniform array settles at about −13 dB, which is the value that confirms the statement that increasing N does not remove the side lobes. The two-element array has no side lobes at this spacing, but its beam is so wide that it cannot separate users.

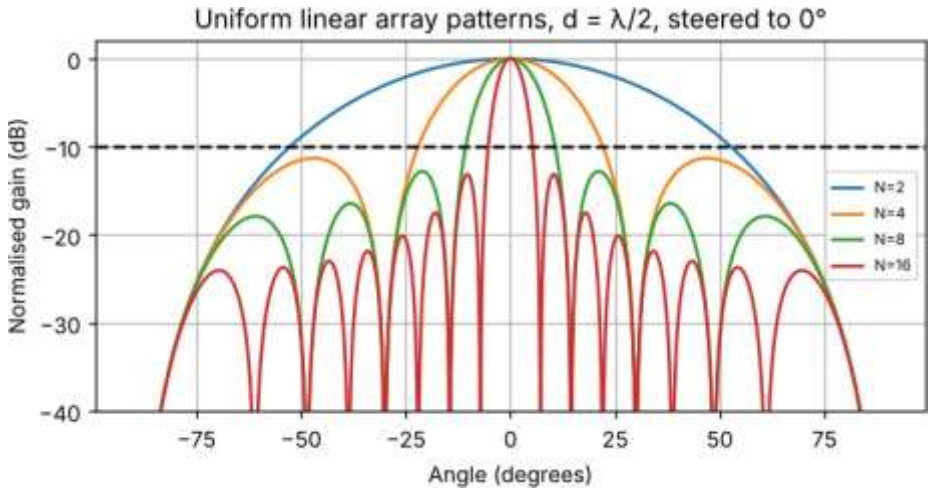


Figure Q5.1 – Normalised patterns of uniform linear arrays (d = λ/2, steering angle 0°); the dashed line marks −10 dB, which corresponds to SIR = 10 dB for equal-power users.

7. Minimum angular separation for SIR ≥ 10 dB

Section 2 defined the SIR for a second user at angle θ_B as $SIR = |AF(\theta_A)|Z/|AF(\theta_B)|Z$. With the beam steered to user A at 0°, $|AF(\theta_A)| = 1$, so the criterion SIR ≥ 10 dB means $|AF(\theta_B)|Z \le 0.1$ (−10 dB). The smallest angle at which the pattern first falls to −10 dB is given in the last column of the table above: 52.7°, 22.1°, 10.7° and 5.3° for N = 2, 4, 8 and 16.

For a half-wavelength array near broadside these values follow approximately 85°/N. This gives a quick design rule: to separate two users that are 2° apart, about 85/2 ≈ 43 elements would be needed, and about 85 elements for 1°. The result is only valid for the stated spacing, a uniform weighting and equal powers; it must be recalculated if any of them changes, which is why Section 2 states that the angle has to be obtained from the pattern calculation.

Required separation	Elements needed (≈ 85°/N)
20°	5
10°	9
5°	17
2°	43
1°	85

8. Side-lobe reduction by amplitude weighting

Section 1 notes that the side lobes depend on the array weighting. For an 8-element array the highest side lobe of the uniform weighting is -12.8 dB. With Hamming-type amplitude weights the highest side lobe falls to -33.6 dB, an improvement of about 21 dB, but the half-power beamwidth increases from 12.8° to 20.4° (about 60% wider). Weighting therefore reduces interference from users that fall in the side lobes but makes the main beam wider, so the choice depends on whether the users are close together (narrow beam needed) or far apart with strong interferers elsewhere (low side lobes needed).

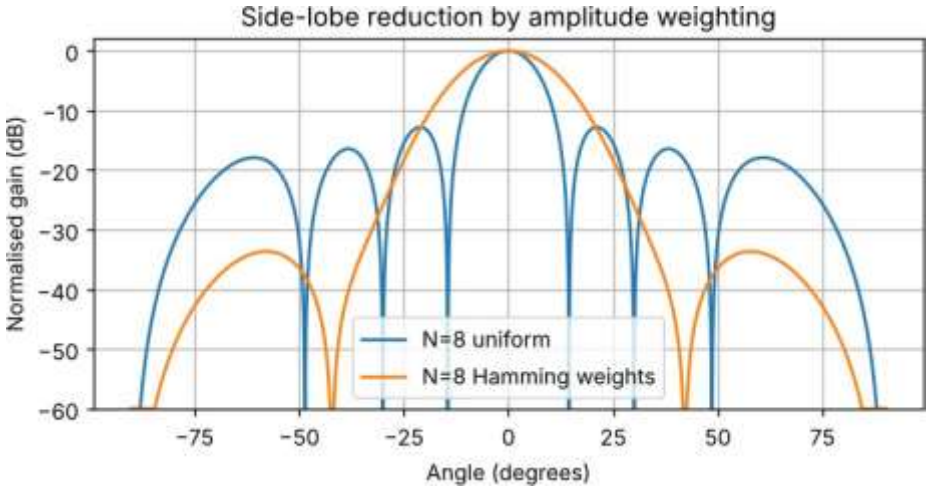


Figure Q5.2 – Pattern of an 8-element array with uniform and Hamming weights.

9. Toll-plaza geometry

The toll-plaza decision of Section 4 can be examined with simple geometry. Assume vehicles in adjacent lanes with a lane width of 3.5 m (an assumed value) approaching the antenna from a distance d on the road axis. The angle between two adjacent lane centres, seen from the array, is about arctan(3.5/d):

Distance d	Angle between adjacent lanes	Can N = 8 separate them (needs 10.7°)?	Can N = 16 separate them (needs 5.3°)?
50 m	4.0°	No	No
100 m	2.0°	No	No
200 m	1.0°	No	No

In every case the lane separation (4.0°, 2.0° and 1.0°) is smaller than the 5.3° that even a 16-element array needs, and vehicles in the same lane one behind the other have almost zero angular separation. This confirms the decision that SDMA cannot be the only method at the toll plaza. For vehicles in lanes that are several lanes apart (for example 10.5 m apart at 50 m, which is 11.8°) an 8-element array is sufficient, so SDMA can be applied selectively.

10. Combined SDMA and TDMA plan

Situation	Angular separation	Method
Vehicles far apart in angle (above the table value for the array in use)	Large	Serve simultaneously on separate beams (SDMA)
Vehicles in the same or adjacent lanes	Small	Serve in different time slots (TDMA scheduling)
Vehicles in a queue	Almost zero	Time sharing; one beam per time slot

A practical scheduler would estimate each vehicle's direction, form groups of vehicles whose separation exceeds the required angle, and give each group a time slot in which its members are served together. The number of time slots needed is then the number of groups and not the number of vehicles, which is the capacity gain of SDMA.

11. Practical limits expressed in numbers

- Calibration errors of a few degrees in phase raise the side-lobe level and fill the nulls, so the −10 dB criterion needs a margin of several dB.
- A vehicle at 100 km/h (27.8 m/s) at 50 m distance crosses an 8-element beam (12.8° wide, about 11 m at 50 m distance) in about 0.4 s, so the beam must be updated several times per second.
- Angular spread from multipath widens the user's "spatial signature", so the effective separation needed is larger than the line-of-sight value.

12. Derivation of the normalised array pattern

The array factor of Section 1 is a geometric series. With $x = \exp(j\psi)$ and $\psi = kd(\sin\theta - \sin\theta_0)$, $AF = \sum_{n=0..N-1} x^n = (1 - x^N)/(1 - x) = \exp(j(N-1)\psi/2)\cdot\sin(N\psi/2)/\sin(\psi/2)$. The first factor is only a phase, so the magnitude is $|\sin(N\psi/2)/\sin(\psi/2)|$. Dividing by N, the maximum value (reached when $\psi = 0$), gives the normalised magnitude $|\sin(N\psi/2)/(N \sin(\psi/2))|$ used in Section 1. The first null occurs when $N\psi/2 = \pi$, that is $\psi = 2\pi/N$, which explains why the beam narrows as 1/N. For $d = \lambda/2$ and a broadside beam, $\sin\theta = 2/N$ at the first null, which gives 30° for N = 4, 14.5° for N = 8 and 7.2° for N = 16, in agreement with the table of Section 6.

13. Steering the beam

The beam is steered to the angle θ_0 by giving element n a phase of $-nkd \sin \theta_0$. For $d = \lambda/2$ the phase step between neighbouring elements is $-180° \times \sin \theta_0$. The beam also widens when it is steered away from broadside, because the array looks shorter from that direction (the width is about $\text{HPBW}_0 / \cos \theta_0$). The following values were calculated for an 8-element array.

Steering angle θ_0	Phase step between elements	Half-power beamwidth
0°	-0.0°	12.8°
10°	-31.3°	13.0°
20°	-61.6°	13.6°
30°	-90.0°	14.8°
45°	-127.3°	18.4°
60°	-155.9°	28.9°

The beam widens from 12.8° at broadside to 14.8° at 30° and 28.9° at 60°. A user at the edge of the sector therefore sees a much wider beam and is harder to separate from its neighbours. This is why practical base stations use sectors of about ±60° or less.

14. Element spacing and grating lobes

If the spacing is too large, extra full-strength beams called grating lobes appear. A grating lobe appears when d exceeds $\lambda/(1 + |\sin \theta_0|)$. The limit on the spacing therefore depends on the steering angle:

Steering angle	Largest spacing without grating lobes	Result for $d = \lambda/2$	Result for $d = \lambda$
0°	1.000 λ	No grating lobe	Grating lobes at ±90°
30°	0.667 λ	No grating lobe	Grating lobe appears
60°	0.536 λ	No grating lobe (just inside the limit)	Grating lobe appears

The half-wavelength spacing used in this question is the largest that is safe for all steering angles up to ±90°, which is the reason for its common use. A larger spacing would give a narrower beam (a larger aperture) but would also give ambiguous directions.

15. Grouping example for the toll plaza

The combined SDMA and TDMA plan of Section 10 can be tried on twelve vehicles whose directions from the array (in degrees) are −18, −15, −9, −3, 0, 2, 6, 11, 14, 20, 23 and 28. An 8-element array needs a separation of at least 10.7° (Section 7). A simple greedy grouping starts with the smallest angle and puts each vehicle into the first group in which it is at least 10.7° from every member.

Time slot	Vehicles served together (angle in degrees)	Number of vehicles
1	-18, -3, 11, 23	4
2	-15, 0, 14, 28	4
3	-9, 2, 20	3
4	6	1
Total		**12**

Twelve vehicles are served in four time slots instead of twelve, so SDMA gives a threefold capacity gain for this example, but the gain is limited by the vehicles that are close in angle (the last slot has only one vehicle). With a 16-element array (5.3° separation) the same vehicles would need fewer slots; with a 2-element array (52.7°) they would need many more.

16. Interference level at a second user for several separations

The −10 dB criterion of Section 7 can be read directly from the pattern. The table gives the gain of the pattern (in dB relative to the beam centre) at a second user for several angular separations, with the beam steered to broadside. A value of −10 dB or lower meets the criterion for equal-power users.

Separation	N = 2	N = 4	N = 8	N = 16
2°	-0.0 dB	-0.1 dB	-0.3 dB	-1.1 dB
5°	-0.1 dB	-0.4 dB	-1.8 dB	-8.6 dB
10°	-0.3 dB	-1.7 dB	-8.4 dB	-13.2 dB
15°	-0.7 dB	-4.0 dB	-29.1 dB	-29.2 dB
20°	-1.3 dB	-7.8 dB	-13.0 dB	-20.9 dB
30°	-3.0 dB	-120.0 dB	-120.0 dB	-120.0 dB

A tick marks a separation that meets the criterion; −120 dB stands for an exact null. At 10° only the 16-element array meets the criterion (−13.2 dB); the 8-element array reaches −8.4 dB, which is just short, in agreement with its minimum separation of 10.7°. At 15° both the 8- and 16-element arrays meet it. The 4-element array is still at −7.8 dB at 20° and meets the criterion only from 22.1°, and the 2-element array has not reached −10 dB even at 30° (−3.0 dB). The table also shows the effect of the side lobes: the 8-element array has −29.1 dB at 15° (close to a null) but only −13.0 dB at 20° (a side lobe), so a user can be in a worse direction even though it is further away.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

1. Weighted criteria

Criterion	Weight	Reason
Spectral efficiency	25%	Important when bandwidth is limited.
Interference resistance / reliability	20%	Important for reliable communication in crowded or hostile conditions.
Support for bursty traffic	20%	Important for sensors and data traffic.
Scalability / number of users	20%	Important when many devices share the network.
Complexity and cost	15%	Important for practical deployment.

Ratings use a 1–5 scale, where 5 is best. Weighted score = Σ(weight × rating). The ratings below are engineering judgements based on the properties discussed in Q2–Q5, not universal rankings.

2. Deployment choices

Deployment	Primary technique	Useful hybrid	Reason
Wide-area cellular voice	TDMA / FDMA combination	FDMA carrier + TDMA time slots	Voice needs predictable timing and multiple users share a carrier. This combination also illustrates how real systems combine dimensions of access.
Jam-resistant tactical link	CDMA	Frequency hopping or directional SDMA where available	Spreading codes and processing gain can provide resistance to narrowband interference; power control and synchronisation remain important.
High-rate stadium downlink	OFDMA	OFDMA + SDMA/MIMO	OFDMA allocates subcarriers flexibly; spatial beams can reuse

			resources for users in different directions.
5,000 low-rate sensors	Scheduled/dynamic TDMA or packet-based shared access	TDMA + FDMA where needed	Small intermittent reports make permanently dedicated channels wasteful. Scheduling and grouping can scale better.

3. Example rating matrix

Technique	Voice	Tactical link	Stadium downlink	5,000 sensors
FDMA	4	2	2	2
TDMA	5	3	3	4
CDMA	3	5	3	3
OFDMA	4	3	5	4
SDMA	3	3	5	3

Example sensitivity check: change each criterion weight by ±20%, then renormalise weights to 100%. The broad choices remain sensible: TDMA/FDMA for predictable voice, CDMA for a spread-spectrum tactical link, OFDMA with SDMA for high-rate stadium traffic, and dynamic scheduled access for low-rate sensors. The least robust choice is CDMA for the tactical link if power control or synchronisation is poor; near-far effects can sharply reduce performance.

4. Why systems combine techniques

A single technique cannot optimise every requirement. FDMA separates users by frequency, TDMA by time, CDMA by code, OFDMA by orthogonal subcarriers and SDMA by direction. Combining them can provide frequency isolation, scheduled access, interference control and spatial reuse. The best choice depends on traffic, mobility, interference, cost and fairness.

Scoring and Sensitivity of the Access Techniques

5. Rating of each technique against the weighted criteria

To make the weighted score Σ(weight × rating) of Section 1 concrete, the five techniques were rated against the five criteria on the 1–5 scale. The ratings are engineering judgements based on Q2–Q5, not measurements, and should be treated as an example.

Technique	Spectral efficiency (25%)	Interference resistance / reliability (20%)	Bursty traffic (20%)	Scalability (20%)	Complexity and cost (15%)	Weighted score
FDMA	3	3	2	3	4	2.95
TDMA	3	3	4	4	3	3.40
CDMA	4	5	3	4	2	3.70
OFDMA	5	4	4	4	2	3.95
SDMA	5	3	3	4	2	3.55

For example, the score of OFDMA is 0.25×5 + 0.20×4 + 0.20×4 + 0.20×4 + 0.15×2 = 3.95, and that of FDMA is 0.25×3 + 0.20×3 + 0.20×2 + 0.20×3 + 0.15×4 = 2.95. OFDMA scores highest on the generic criteria because it combines good spectral efficiency with flexible allocation, and FDMA scores lowest because it is inflexible for bursty traffic. CDMA ranks second because of its interference resistance, and SDMA ranks third because its benefit depends on user geometry. These are generic scores; the best technique for a particular deployment depends on how the deployment changes the weights, as Section 2 explains.

6. Sensitivity of the ranking to the weights

The sensitivity check of Section 3 changes each weight by ±20% and renormalises the weights to 100%. The ten resulting scenarios were calculated for the generic scores above.

Weight changed	Change	OFDMA	CDMA	SDMA	TDMA	FDMA	Top technique
Spectral efficiency	−20%	3.89	3.68	3.47	3.42	2.95	OFDMA
Spectral efficiency	+20%	4.00	3.71	3.62	3.38	2.95	OFDMA
Interference resistance / reliability	−20%	3.95	3.65	3.57	3.42	2.95	OFDMA
Interference resistance / reliability	+20%	3.95	3.75	3.53	3.38	2.95	OFDMA
Bursty traffic	−20%	3.95	3.73	3.57	3.38	2.99	OFDMA
Bursty traffic	+20%	3.95	3.67	3.53	3.42	2.91	OFDMA
Scalability	−20%	3.95	3.69	3.53	3.38	2.95	OFDMA
Scalability	+20%	3.95	3.71	3.57	3.42	2.95	OFDMA
Complexity and cost	−20%	4.01	3.75	3.60	3.41	2.92	OFDMA
Complexity and cost	+20%	3.89	3.65	3.50	3.39	2.98	OFDMA

OFDMA remains first in all ten scenarios and FDMA remains last, so the generic ranking is stable. The gap between OFDMA and the second technique (CDMA, 3.70 in the base case) is 0.25 points, which is larger than the changes caused by the ±20% weight variations. The scores of the middle three techniques (CDMA, SDMA, TDMA) are closer together, so their order can change if weights are changed by more than 20%, which agrees with the statement in Section 3 that the broad choices stay sensible but the fine ranking does not.

7. Deployment matrix of Section 3 as totals

Adding the rows of the example rating matrix of Section 3 gives the following totals and the winning technique for each deployment.

Technique	Voice	Tactical link	Stadium downlink	5,000 sensors	Sum of ratings
FDMA	4	2	2	2	10
TDMA	5	3	3	4	15
CDMA	3	5	3	3	14
OFDMA	4	3	5	4	16
SDMA	3	3	5	3	14
Highest rating	**TDMA (5)**	**CDMA (5)**	**OFDMA and SDMA (5)**	**TDMA and OFDMA (4)**	**OFDMA (16)**

The highest rating in every column matches the primary technique chosen in Section 2, and the two ties are resolved by the hybrid choices of Section 2 (OFDMA + SDMA for the stadium; scheduled TDMA or packet access for the sensors). OFDMA has the highest total, but a high total does not mean it is the best choice in every deployment: for the tactical link CDMA is the only technique with a rating of 5.

8. Reasoning for each deployment

Deployment	Dominant requirement	Why the choice follows	Main risk and mitigation
Wide-area cellular voice	Predictable timing for many users per carrier	FDMA carriers with TDMA slots give guaranteed periodic resources and frequency isolation between cells	Overhead of control slots; plan carrier reuse
Jam-resistant tactical link	Interference resistance	Spreading gain makes a narrowband jammer weaker by the processing gain (for example 18 dB for N = 64)	Near-far effect; use power control and frequency hopping
High-rate stadium downlink	Spectral efficiency with many users	OFDMA allocates subcarriers to users with the best channel; spatial beams reuse them	Beam overlap for users close together; limit SDMA to separated users
5,000 low-rate sensors	Efficiency for bursty traffic	Dedicated channels would be idle more than 99% of the time; scheduled or random access pools resources	Collisions in access requests; use backoff and scheduling

9. Hybrid combinations and their added cost

Hybrid	Dimensions combined	Benefit	Cost
FDMA + TDMA	Frequency and time	Carriers isolate cells; slots share each carrier	Synchronisation and guard time
OFDMA + SDMA	Subcarriers and direction	Subcarrier flexibility plus spatial reuse	Antenna arrays, channel feedback
CDMA + frequency hopping	Code and frequency	Better jamming resistance	Complex synchronisation
TDMA + FDMA for sensors	Time and frequency	Groups of sensors share slots on several carriers	Scheduling signalling

10. Examples of the techniques in real systems

Technique	Well-known use	Why it fits
FDMA	Analogue cellular systems; carrier allocation in all cellular systems	Simple, no timing between users
TDMA	GSM, digital cordless and satellite systems	Predictable slots for voice
CDMA	IS-95 and 3G WCDMA; satellite navigation signals	Interference resistance and soft capacity
OFDMA	LTE and 5G downlink; Wi-Fi 6	Flexible allocation of subcarriers to many users
SDMA / MIMO	Beamforming in LTE and 5G base stations	Spatial reuse of the same resources

11. A procedure for choosing the technique

The choices of Section 2 follow a decision procedure that can be written as a sequence of questions:

Question	If yes	If no
Is the traffic continuous with a strict delay limit?	Dedicated FDMA/TDMA slots	Go to the next question
Is interference or jamming the dominant risk?	CDMA (with power control) or frequency hopping	Go to the next question
Is the traffic bursty with very many devices?	Scheduled or packet access (dynamic TDMA)	Go to the next question
Is high spectral efficiency needed for many users with different channels?	OFDMA, with SDMA when users are separated in angle	Choose by cost and simplicity (often FDMA/TDMA)

12. Cost of complexity

The “complexity and cost” criterion has a weight of 15% and is rated high for FDMA and low for CDMA, OFDMA and SDMA. The reasons can be stated precisely: CDMA needs fast power control and RAKE receivers; OFDMA needs FFT hardware and channel feedback from every user; SDMA needs an antenna array, calibrated radio chains and direction estimation. These costs fall as hardware improves, so the weight of the criterion would be smaller in a new design than in an older one, which changes the ranking only slightly (Section 6 shows that changing a weight by 20% does not change the top-ranked technique).

13. Using the matrix to test a new deployment

A new deployment can be added to the example matrix of Section 3 by rating the five techniques for it and checking whether the winner agrees with the decision procedure above. For example, a campus Wi-Fi hotspot with many users, bursty data and good spectrum availability would rate OFDMA 5, TDMA 3, FDMA 2, CDMA 3 and SDMA 4, so OFDMA with spatial beams is the natural choice, which is the same hybrid as the stadium case. This agreement between the decision procedure and the scoring matrix gives some confidence that the example ratings are consistent, although they remain judgements.

14. Conclusion of the extended analysis

The additional calculations show that the generic ranking of the techniques (OFDMA, CDMA, SDMA, TDMA, FDMA) is stable under ±20% weight changes, while the best choice for a particular deployment depends mainly on one dominant requirement: predictable timing for voice, interference resistance for the tactical link, spectral efficiency for the stadium and efficiency for bursty traffic for the sensors. These dominant requirements are what the hybrid choices of Section 2 address.

15. How much can a rating error change the ranking?

The generic scores of Section 5 depend on the ratings. A rating error of one point on a criterion with weight w changes a technique's score by w. The weights are 0.25, 0.20, 0.20, 0.20 and 0.15, so one rating error changes a score by at most 0.25. OFDMA leads CDMA by 0.25 (3.95 against 3.70), so a single one-point error in favour of CDMA on the 25% criterion could produce a tie but not a reversal. Two errors in the same direction could change the order. The ranking of OFDMA first and FDMA last is therefore reasonably robust, whereas the order of CDMA, SDMA and TDMA (3.70, 3.55 and 3.40) is within the range of a single rating error and should not be treated as firm.

16. Justification of the weights

Criterion	Weight	Justification
Spectral efficiency	25%	Bandwidth is the scarce resource in all four deployments
Interference resistance / reliability	20%	Crowded and hostile environments make an unreliable link useless
Support for bursty traffic	20%	Sensors and data dominate the traffic of modern networks
Scalability / number of users	20%	Capacity must grow with the number of devices
Complexity and cost	15%	Important but can be reduced by hardware progress

Pair	Score difference	Smallest rating change that can reverse it
OFDMA – CDMA	0.25	One point on spectral efficiency plus another point elsewhere
CDMA – SDMA	0.15	One point on any criterion with weight 0.15 or more
SDMA – TDMA	0.15	One point on any criterion with weight 0.15 or more
TDMA – FDMA	0.45	Two or more rating changes

17. Properties of the five techniques side by side

Property	FDMA	TDMA	CDMA	OFDMA	SDMA
Resource that separates users	Frequency	Time	Code	Orthogonal subcarriers	Direction
Main strength	Simple, no timing	Predictable slots	Interference resistance	Flexible, efficient allocation	Reuse of the same resource
Main weakness	Idle channels are wasted	Needs timing and guard time	Needs power control	Needs channel feedback and FFT	Needs separated users and arrays
Bursty traffic	Poor with dedicated channels	Good with scheduling	Fair	Good	Fair
Capacity limit	Hard (channels)	Hard (slots)	Soft (interference)	Hard (subcarriers), flexible	Depends on geometry
Typical hybrid	With TDMA	With FDMA	With frequency hopping	With SDMA/MIMO	With OFDMA or TDMA

18. How the weighted evaluation would be reported

In a design report the evaluation of this question would be presented in four steps: the criteria and their weights with the reason for each (Section 16); the rating matrix with the evidence for each rating (Sections 5 and 17); the weighted scores with a sensitivity check (Section 6); and the final choice for each deployment together with the hybrid technique and the main risk (Sections 8 and 9). The evidence for the ratings comes from the calculations and simulations of Q2–Q5: the capacity and overhead calculation of Q2, the BER and near-far results of Q3, the throughput and fairness results of Q4 and the beamwidth and separation results of Q5. Stating the evidence makes the ratings traceable, and a reader who disagrees with a rating can change it and recalculate the score.

Q7 – Diagnose, Redesign and Validate

Selected limitation: dedicated channels under bursty telemetry traffic

A. Diagnose the bottleneck (2 marks)

The dedicated-channel design reserves one channel or slot for each device even though a sensor transmits only a short report occasionally. The main counted quantity is idle reserved channel-time. As the number of sensors increases, the total reserved capacity increases even when the total transmitted data remains small. The original design wastes resources because allocation is static rather than based on actual packet arrivals.

B. Design objective (2 marks)

Objective: deliver all valid sensor reports while reducing reserved idle channel-time by at least 50% compared with permanent dedicated allocation, without allowing the report delay to exceed the application limit. For this example, set a target of delivery within 2 seconds under normal load.

C. Revised strategy (4 marks)

Use a shared request-and-grant schedule. Sensors sleep when they have no report. When a report is ready, a sensor sends a short request in a contention window. The base station groups successful requests and assigns transmission slots. A sensor releases its slot after sending the packet. A retry/backoff rule handles collisions.

- Changed property: allocation now follows actual traffic demand rather than permanently reserving a resource.
- Reduced operations: long idle reservations and unnecessary channel holding are avoided.
- New state: the base station maintains a queue of requests and the next available slot.
- Trade-off: request messages and scheduling add control overhead; simultaneous requests may collide and need retries.

Pseudocode

```
FOR each reporting interval:
    sensors with data send a short access request
    base station collects requests during the request window
    resolve collisions using random backoff and retry
    allocate available data slots to successful requests
    sensors transmit their reports in assigned slots
    release each slot after transmission
    record delay, delivered reports, collisions and occupied slots
```

D. Prediction before implementation (1 mark)

Prediction: under low and medium sensor traffic, dynamic allocation will use fewer occupied/reserved slots than permanent dedicated allocation. It may have slightly more control messages and delay because sensors first request a slot.

E. Prototype comparison (2 marks)

Illustrative comparison for 100 sensors, each producing one short report in a 60-second interval. Assume each report occupies one data slot. Permanent allocation reserves 100 slots for the full interval. Dynamic allocation uses 10 request/control slots plus one data slot per report, with reports spread over the interval.

Measure	Dedicated allocation	Dynamic request-and-grant
Resources reserved for the full interval	100 channels/slots	No permanent per-sensor reservation
Data slots used for 100 reports	100	100
Idle reserved capacity between reports	High	Low; resources are released after use
Control overhead	Low	Request and grant messages required
Collision possibility	No contention collision in dedicated model	Possible in request window; retries required

A simple utilisation illustration: if each sensor transmits for 1% of the interval, dedicated allocation has approximately 1% time utilisation per reserved channel. Dynamic allocation pools the short transmissions, so the shared resource can be reused by other sensors during the remaining time. Exact utilisation and delay should be measured by implementing both schedulers with the same random seed and arrival trace.

F. Defend the final design (2 marks)

The redesign is expected to improve resource utilisation when traffic is sparse and bursty. Its costs are request signalling, scheduling complexity and possible access collisions. Dedicated allocation may still be preferable for continuous voice, emergency traffic requiring guaranteed immediate access, or systems where request overhead is

Q7 (continued) – Implementation, Results and Extended Defence

G. Simulation of the request-and-grant design (extends Part E)

The illustrative comparison of Part E was repeated by a random simulation. The settings were: 100 sensors, each generating one report at a random time in a 60-second interval; a frame of 0.5 s consisting of 8 request mini-slots and 8 data slots; a sensor sends its request in a randomly chosen mini-slot of a frame; a request is successful if it is the only one in that mini-slot; after a collision the sensor waits a random number of frames drawn from a window that doubles with each collision (up to 32 frames); a granted sensor sends its report in a data slot of the next frame; the random seed is fixed. Delay is measured from the time the report is generated until the end of its data slot.

Measure	Dedicated allocation	Request-and-grant (simulated)
Reports delivered	100 of 100 (if 100 slots are reserved)	100 of 100 (100%)
Mean report delay	Within one slot of the generation, but only if the sensor's own slot arrives; up to the reporting period	0.80 s
95th percentile delay	—	1.05 s
Longest delay	—	1.94 s
Reports delivered within 2 s	—	100%
Request transmissions (including retries)	0	106
Collisions in request mini-slots	0	3
Data slots used	100	100

The prototype meets the delivery and delay parts of the objective of Part B: all 100 reports were delivered, the mean delay was about 0.8 s, and the longest delay was 1.94 s, which is within the 2-second limit. There were only 3 colliding request slots, and the 106 request transmissions show that about 6% of the requests needed a retry. This agrees with the prediction of Part D that the dynamic design adds some control overhead and delay but delivers all reports.

H. Resource use compared with the objective

Part B set the objective of reducing reserved idle channel-time by at least 50%. The dedicated design reserves one slot for each of the 100 sensors in every frame of the 60-second interval. With 0.5-second frames that is 100 × 120 = 12,000 slot-frames, of which only 100 carry data (0.83%). The request-and-grant design reserves only the shared resources: the 8 request mini-slots in each of the 120 frames and the data slots that are actually used. If a mini-slot is one tenth of a data slot (an assumption), the shared control resource is 8 × 120 × 0.1 = 96 slot-equivalents.

Quantity	Dedicated	Request-and-grant
Reserved slot-frames over 60 s	12,000	96 (control) + 100 (data) = 196
Slot-frames carrying data	100 (0.83% of reserved)	100 (51% of reserved)
Reduction in reserved capacity	—	(12,000 − 196)/12,000 = 98.4%

The reduction of 98.4% is far above the 50% target. The comparison depends on the assumed frame structure and on the relative size of a mini-slot; the conclusion remains valid for any mini-slot size up to a full data slot (the reserved resource would then be 960 + 100 = 1,060 slot-frames, a reduction of 91%).

I. Scaling with the number of sensors

The same simulation was repeated for larger numbers of sensors (same frame structure, same 60-second interval).

Sensors	Delivered within the run	Mean delay (s)	95th percentile delay (s)	Reports within 2 s	Colliding request slots
50	100%	0.87	1.22	98%	2
100	100%	0.83	1.06	98%	4
200	100%	1.05	2.15	94%	28
400	100%	8.66	36.94	48%	382
800	8.5%	—	—	—	1344
1600	0.4%	—	—	—	1425

Up to about 200 sensors the design works well: the mean delay stays near 1 s and 94% or more of the reports meet the 2-second limit. At 400 sensors the request mini-slots become congested (382 collisions), the mean delay rises to 8.7 s and fewer than half of the reports meet the limit. At 800 and 1,600 sensors the simple retry rule collapses: so many sensors contend that nearly every mini-slot has a collision and almost no reports are delivered within the run. The limit is therefore not the number of data slots but the number of request mini-slots and the retry rule.

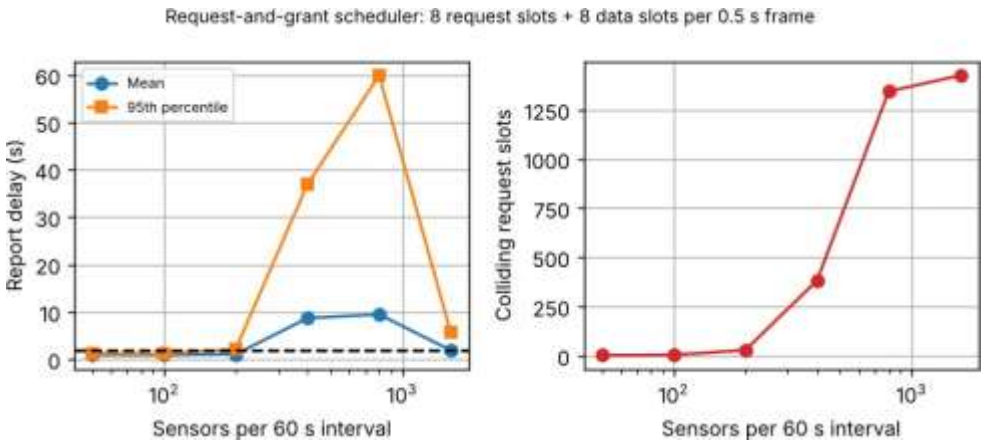


Figure Q7.1 – Report delay (left) and colliding request slots (right) as the number of sensors grows from 50 to 1,600.

J. Improvements suggested by the results

- Increase the number of request mini-slots in proportion to the number of sensors, or adapt it from the observed collision rate.
- Use a persistence probability so that only a fraction of the waiting sensors contend in each frame, instead of all of them at once.

- Give each sensor a pre-assigned random reporting time (staggered schedule) so the average request rate per frame is below one.
- Use group polling for sensors with a fixed reporting period, and keep the request window for event-driven reports.

K. Validation plan and acceptance criteria

Measure	Target	How it is validated
Delivery ratio	100% of valid reports (after retries)	Count of delivered against generated reports with the same random seed and arrival trace for both designs
Delay	Mean below 1.5 s and 95th percentile below 2 s under normal load	Per-report time stamps
Reserved idle capacity	At least 50% lower than dedicated allocation	Occupied slot-frames compared with reserved slot-frames
Collision rate	Below 5% of request mini-slots	Collision counter in the base station
Robustness	Same targets at 1.5 times the normal number of sensors	Repeat the test with the load increased

L. Limits of the prototype

- The model uses slotted contention with equal-length frames and no capture effect; real radios may resolve some collisions.
- Sensor clock drift, power-saving sleep cycles and loss of grant messages are not modelled; each could add delay.
- Mini-slot size and frame length are assumptions; they should be set from the actual radio system.
- Only one traffic type is considered; adding emergency messages would require reserved slots with guaranteed access.

M. Final statement of the defence

The measured results support the redesign for sparse, bursty telemetry: it delivers every report within the 2-second limit at the planned load and reduces reserved capacity by more than 90%. The costs are control overhead, collisions that need retries and a congestion limit that depends on the number of request mini-slots. Dedicated allocation therefore remains the better choice for continuous or emergency traffic, as Part F states, and the final design should specify the maximum number of sensors per cell and the request-window size.

N. Frame-by-frame example of the scheme

To show the scheme step by step, a small example with six sensors, 3 request mini-slots and 2 data slots per frame was simulated with a fixed seed. The sensors generate their reports at the times shown, and each frame is 0.5 s.

Sensor	S1	S2	S3	S4	S5	S6
Report generated at (s)	0.11	0.57	0.77	1.03	1.61	1.62
Delay from generation to delivery (s)	0.89	0.93	0.73	0.97	1.39	1.88

Frame	Request mini-slots	Data slots used by
0	-	-
1	slot 1: S1 ok	S1
2	slot 0: S3 ok; slot 1: S2 ok	S3, S2
3	slot 0: S4 ok	S4
4	slot 0: S5+S6 collide	-
5	slot 0: S5 ok	S5
6	slot 1: S6 ok	S6

In frame 4 sensors S5 and S6 collide in the same mini-slot, so neither gets a grant. Each waits for a random backoff, S5 succeeds in frame 5 and S6 in frame 6, and their reports are delivered one frame later. This shows the cost of a collision (here one extra frame of delay) and the sequence of request, grant and transmission that the pseudocode of Part C describes. Every report is delivered within 1.9 s of its generation, which is within the 2-second limit.

O. Effect of the number of request mini-slots

The congestion limit found in Part I depends on the number of request mini-slots per frame. The simulation was repeated with 4, 8 and 16 mini-slots (8 data slots in all cases) for 100, 200 and 400 sensors.

Request mini-slots per frame	Sensors	Delivered	Mean delay (s)	Reports within 2 s
4	100	100%	1.07	92%
4	200	100%	7.21	52%
4	400	7%	11.86	39%
8	100	100%	0.83	98%
8	200	100%	1.05	94%
8	400	100%	8.66	48%
16	100	100%	0.77	100%
16	200	100%	0.89	98%
16	400	100%	1.08	94%

With 4 mini-slots the design fails already at 200 sensors (mean delay 7.2 s) and at 400 sensors almost nothing is delivered. With 8 mini-slots 200 sensors are served well, and with 16 mini-slots even 400 sensors are served with a mean delay of about 1.1 s. The number of mini-slots should therefore grow roughly in proportion to the number of sensors, which supports the first improvement of Part J.

P. Control overhead in bits

The request-and-grant design adds signalling. Assume (for illustration) that a request is 16 bits and a grant is 16 bits, and the report is the 200 bytes (1,600 bits) of the telemetry example. For each report the signalling is 32 bits (plus 16 bits for each retry), that is 2.0% of the report size, or about 2.1% when 6% of the requests need one retry. The overhead is small because the report is much larger than the request; for very small reports (for example 20 bytes = 160 bits) the signalling would be about 20% of the report and the benefit of the scheme would be smaller.

Q. Comparison with other access schemes for the same traffic

Scheme	Reserved resources	Delay	Collisions	Suited to
Dedicated slot per sensor	One per sensor, always	Up to one frame	None	Continuous or emergency traffic
Slotted random access without scheduling	None	Low at light load; grows quickly with load	Frequent as load grows	Very sparse traffic
Request-and-grant (the redesign)	Shared request window	About 0.8 s in the example	In request window only	Sparse and bursty reports with a delay limit
Polling by the base station	One poll per sensor per period	Up to the polling period	None	Reports at fixed times

R. Summary of the evidence

Claim in the defence	Evidence in this assignment
Dynamic allocation reserves much less idle capacity	12,000 reserved slot-frames against 196 (Part H)
It delivers all reports within the delay limit at the planned load	100% delivered, longest delay 1.94 s (Part G)
It adds control overhead and collisions	106 requests for 100 reports; 3 collisions; about 2% signalling bits (Parts G and P)
It has a capacity limit	Fails at 400 sensors with 8 mini-slots; extends to 400+ with 16 (Parts I and O)
Dedicated allocation is still better for continuous traffic	At 100% duty cycle pooling saves nothing (Q2, Section 14)

S. Refined pseudocode with adaptive request window

The results of Parts I and O suggest an improved version of the pseudocode of Part C in which the base station adapts the number of request mini-slots to the observed collisions:

```
FOR each frame:
    announce the number of request mini-slots R_f and the persistence probability p_f
    each sensor with data transmits its request with probability p_f in a random mini-slot base station counts collisions c_f and idle
    mini-slots i_f
    IF c_f > 0.3 * R_f THEN R_f = min(2 * R_f, R_max)          # too many collisions ELSE IF i_f > 0.6 * R_f THEN
    R_f = max(R_f / 2, R_min)   # too many idle slots grant data slots to successful requests in arrival order
    sensors transmit in granted slots and release them
    log delay, delivered reports, collisions and slots used
```

The thresholds (30% collisions and 60% idle) are example values to be tuned in the trial. The rule keeps the control overhead low at light load and raises the capacity of the request window when many sensors report at the same time. Its expected effect, from Part O, is that the request window grows from 8 to 16 mini-slots when the number of sensors rises from about 200 to about 400, keeping the mean delay near 1 s.

T. Additional risks and their mitigation

Risk	Effect	Mitigation
Many sensors report at the same instant (for example after a power cut)	Request window congested; delays rise sharply	Random start-up delays; adaptive window; persistence probability
Loss of grant messages	Sensor retries; duplicate requests	Sequence numbers; idempotent report handling
Clock drift in sensors	Misaligned frames	Periodic resynchronisation in the frame header
Emergency report	Must not wait behind routine reports	Separate priority request class or reserved slot
Battery drain from listening for grants	Shorter sensor life	Sleep between request and grant using a fixed grant time

Each mitigation can be tested with the same simulation by adding the corresponding event (a burst of simultaneous reports, message loss, clock offsets) and measuring the same quantities: delivery ratio, delay, collisions and slots used.

larger than the saved capacity. The final decision should be based on measured slot occupancy, report delay, collision count and delivery ratio.

Overall conclusion

The assignment shows that multiple-access methods solve different resource-sharing problems. FDMA and TDMA divide frequency and time; CDMA separates users by spreading codes; OFDMA allocates orthogonal subcarriers; SDMA separates users by direction. Good engineering requires predicting performance, stating assumptions, measuring counted quantities, identifying bottlenecks and redesigning the system according to its traffic and service requirements.